AI-Driven Multi-Antenna Designs for Next Generations of Wireless Communication Systems

Doctoral Thesis Defense

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September 25, 2025



Outline

Introduction

Part I - Channel Estimation

Part II - Symbol Detection

Part III - Precoder Optimization

Conclusions

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- To address the growing demands, advanced technologies like massive multi-input and multi-output (mMIMO), millimeter wave (mmWave) communications, intelligent reflecting surfaces, etc. has led to increasing system complexity.
- Designing wireless communication systems.
  - Conventional approaches: domain knowledge, mathematical models, optimization algorithms.
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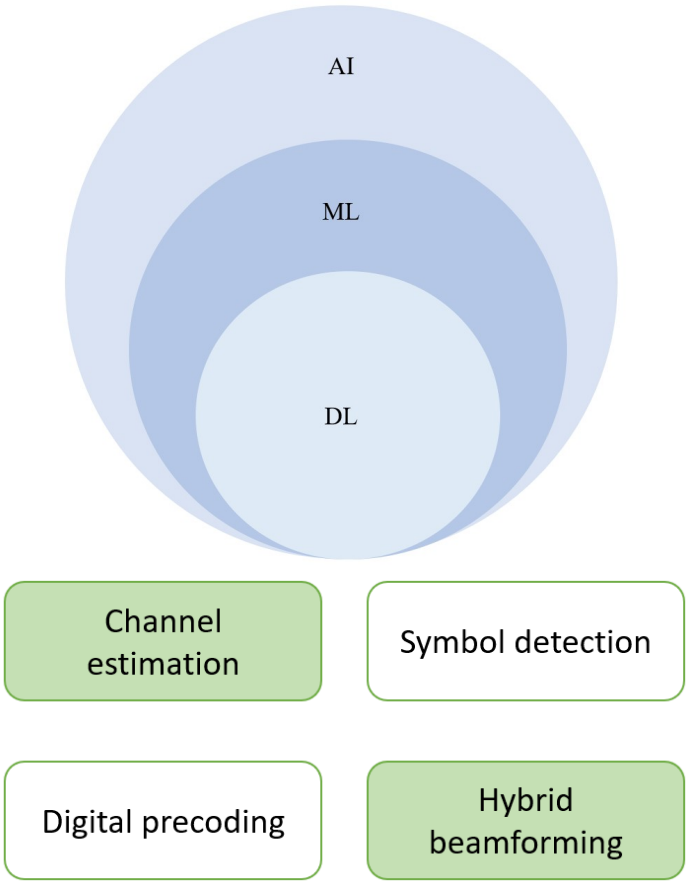
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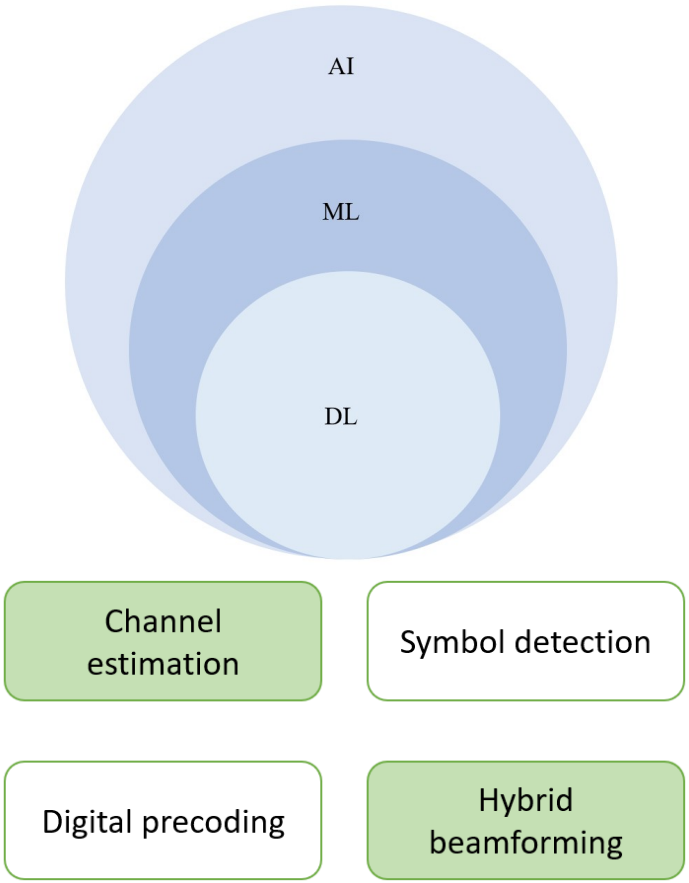
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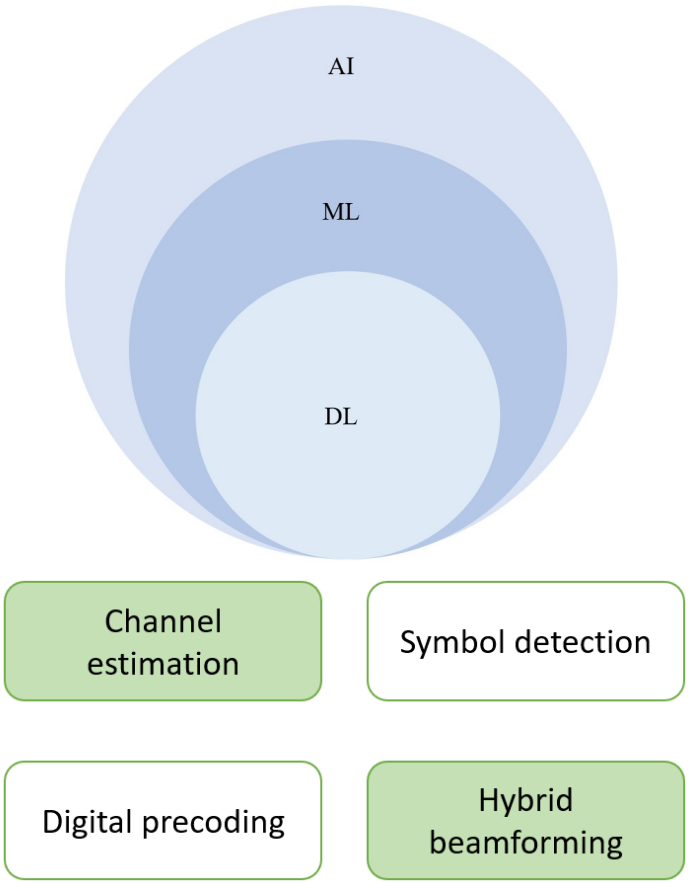
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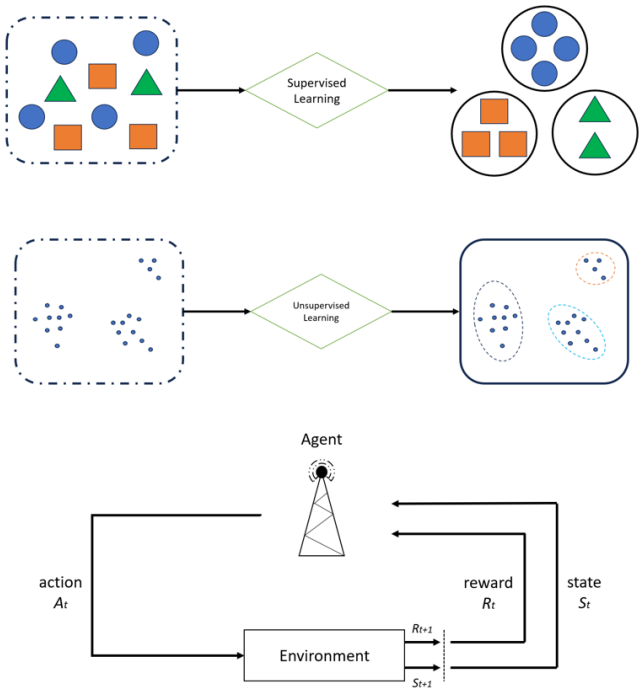
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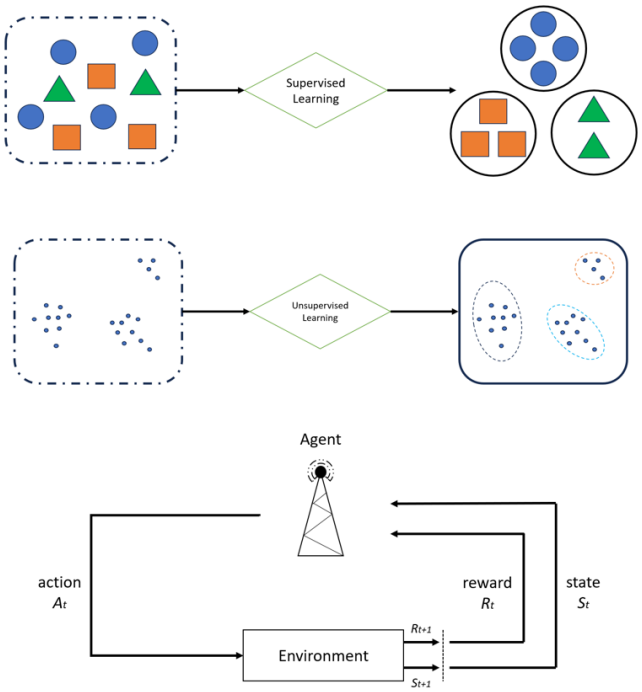
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- Rapid evolution of AI over the past few decades.
- Levels of supervision:
  - Supervised learning (SL)
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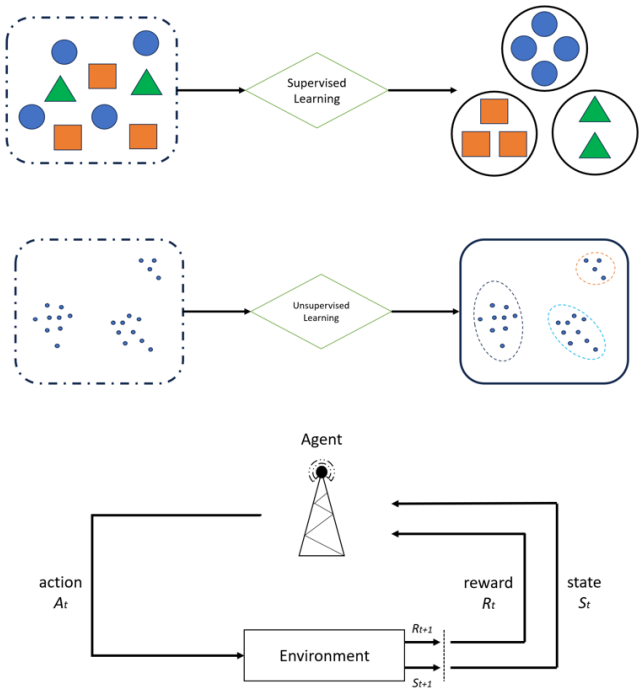
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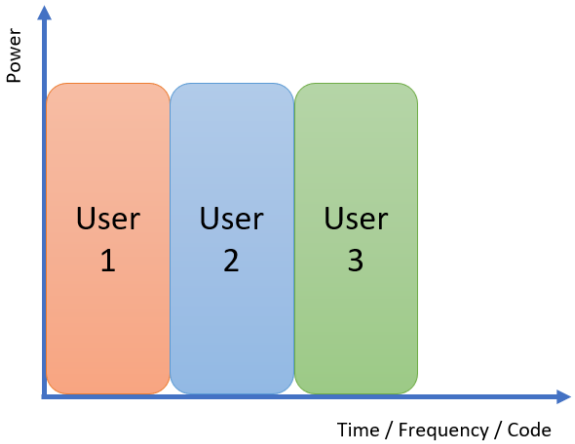
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Multiple Access Schemes

- With the increase in different types of devices with varying requirements, there is a need for advanced multiple access schemes to manage multi-user interference.
- Orthogonal Multiple Access (OMA) allocates orthogonal radio resources.
- Space Division Multiple Access (SDMA) involves transmit side interference cancellation by spatial precoding and fully treating interference as noise at receivers.



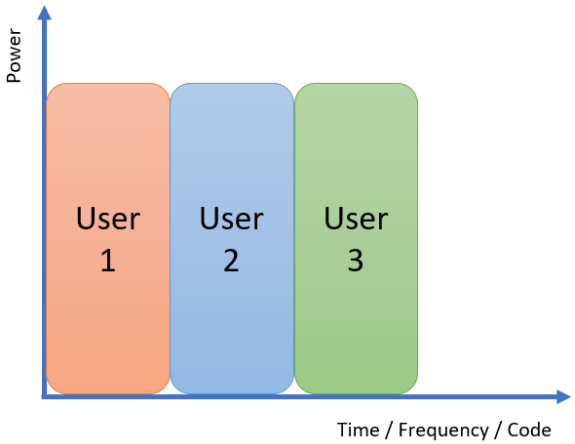
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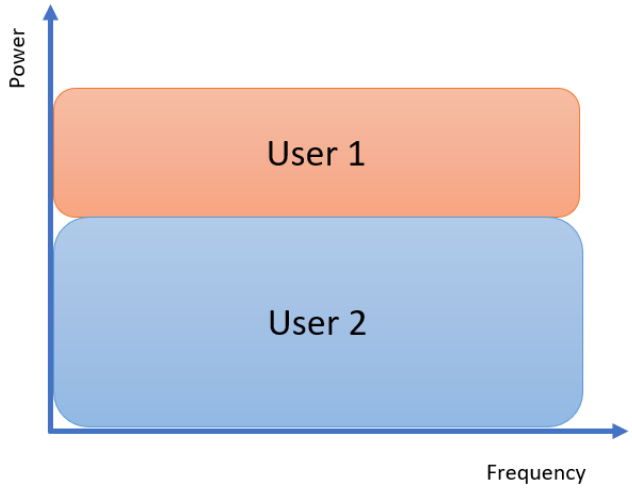
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SDMA

NOMA

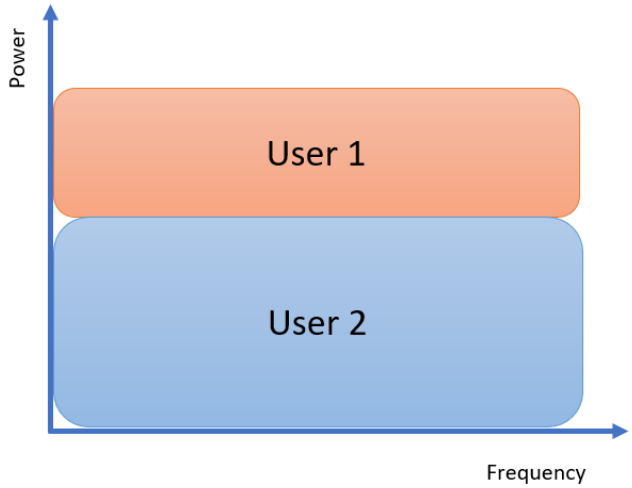
- Non-orthogonal multiple access (NOMA)
  - Use power domain to break orthogonality.
  - Apply superposition coding (SC) at the transmitter and successive interference cancellation (SIC) at each receiver.
- Management of interference
  - OMA: no interference.
  - SDMA: treat interference as noise.
  - NOMA: fully decode interference.



PD-NOMA

NOMA

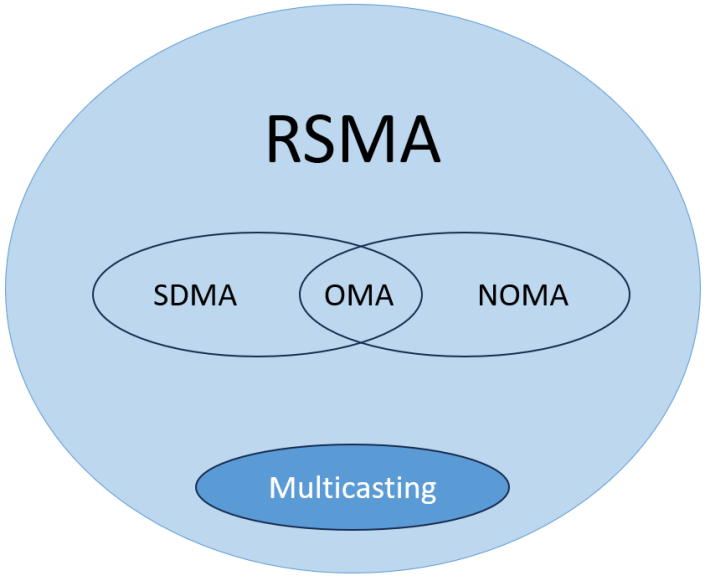
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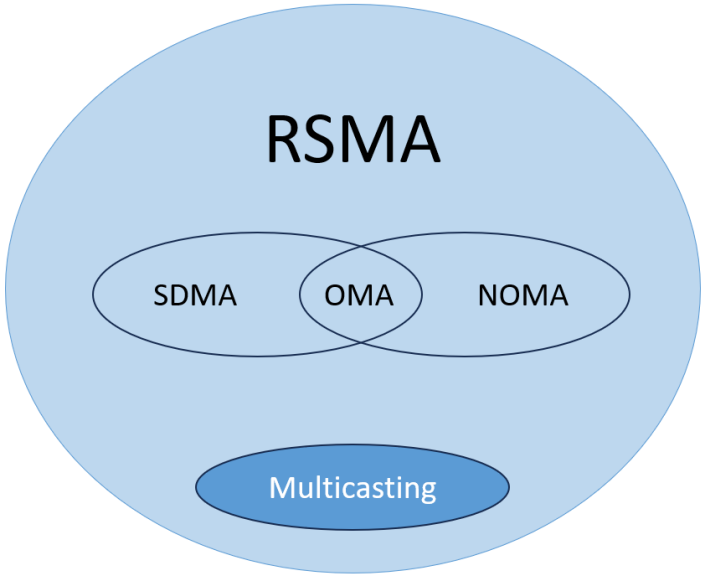
RSMA

- Rate-Splitting Multiple Access (RSMA)
  - Partially treat interference as noise, partially decode.
- Involves both transmit-side and receive-side interference cancellation.
- RSMA encompasses OMA, SDMA, NOMA, multicasting.
- Has been shown to be robust to CSI uncertainty, flexible to different user deployments.



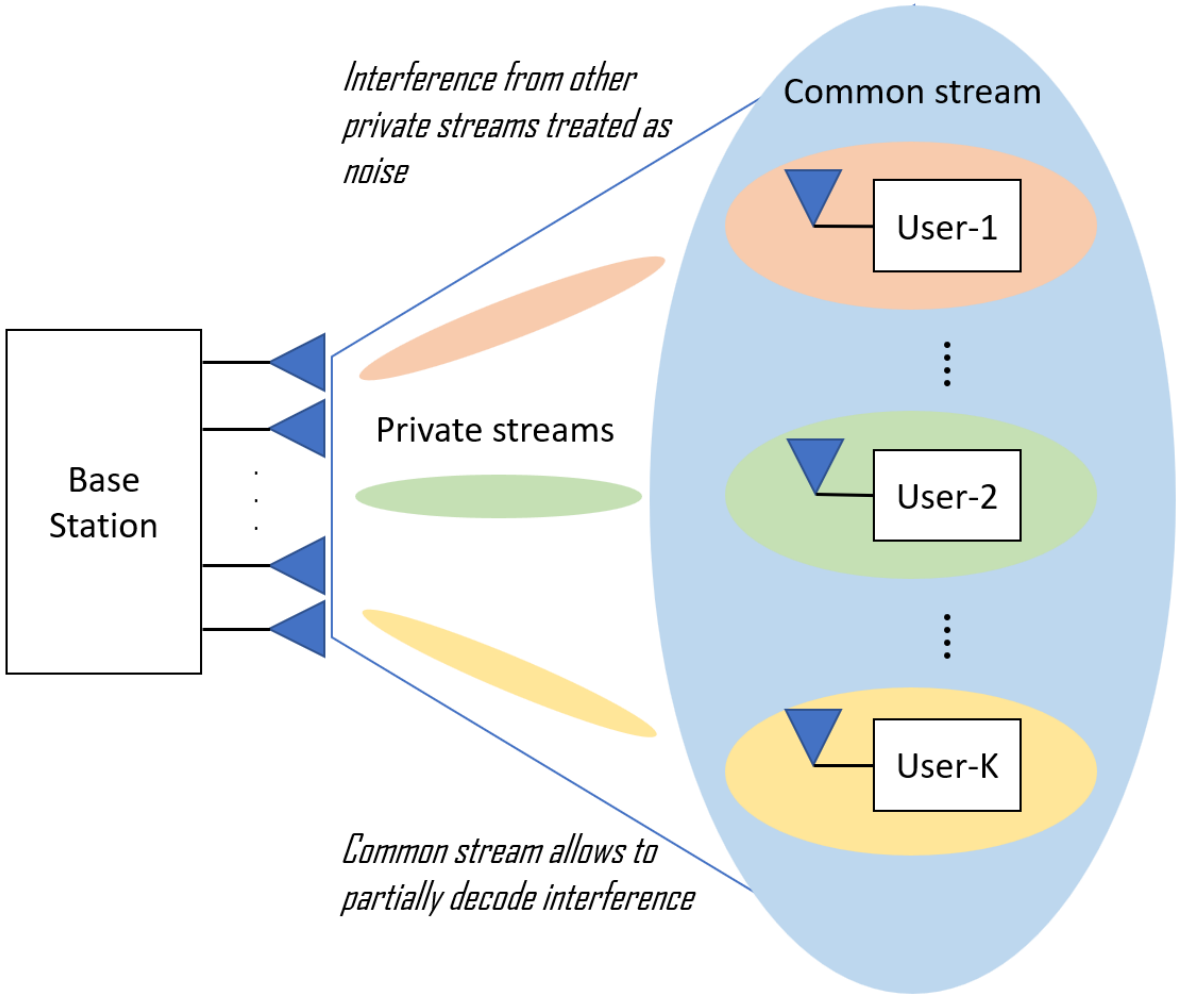
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RSMA

- Simple visualization of 1-layer RSMA



Thesis Objectives

Design and application of AI/ML techniques

1. Retrieve CSI with high precision
   - Conventional channel estimation schemes offer limited performance in large MIMO settings
2. Robust symbol detection in RSMA
   - Sensitivity of conventional receivers to CSI and improper SIC
   - Linear and non-linear channel conditions
3. Data-driven digital precoding scheme
   - High computational complexity of WMMSE precoding scheme
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   - Extension of precoding to hybrid beamforming

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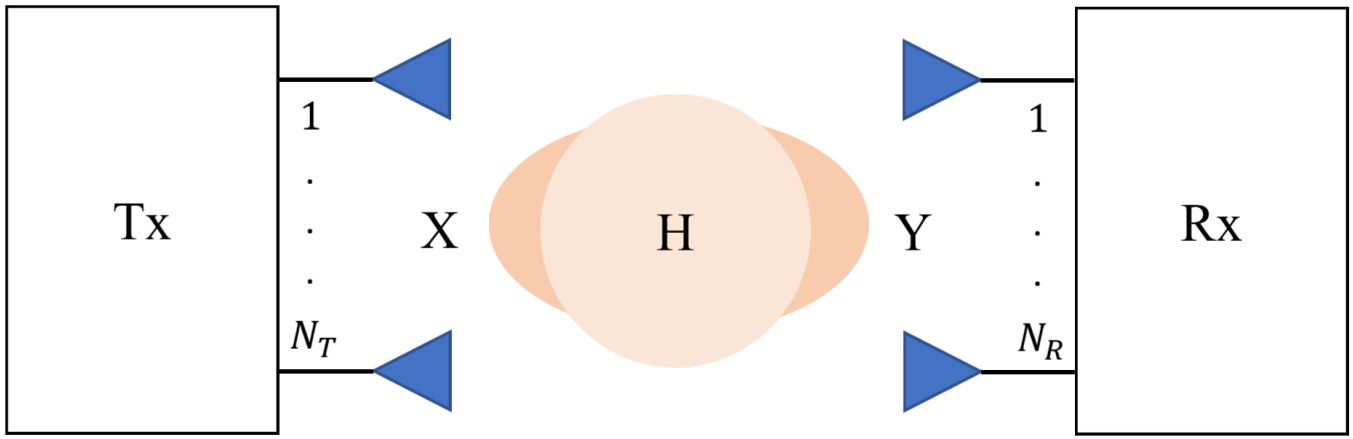
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Motivation

- Obtaining reliable channel state information (CSI) is crucial for signal processing tasks.
  - Linear schemes often used: linear least-squares (LS) and minimum mean squared error (MMSE).
  - Existing machine learning approaches: high computational complexity during training and inference phases.



Signal Model

Received signal when transmitting N_P pilot symbols in an $N_{\mathrm{T}} \times N_{\mathrm{R}}$ point-to-point MIMO system:

$$\mathbf{Y} = \mathbf{H}\mathbf{X} + \mathbf{N}$$

- $\mathbf{Y} = [\mathbf{y}_1, \ldots, \mathbf{y}_{N_P}] \in \mathbb{C}^{N_R \times N_P}$
- $\mathbf{H} = \mathbf{C}_{\mathrm{Rx}}^{1/2}\, \mathbf{G}\, \mathbf{C}_{\mathrm{Tx}}^{1/2} \in \mathbb{C}^{N_{\mathrm{R}} \times N_{\mathrm{T}}}$
  - $\mathbf{G}$ modeled as i.i.d. circularly symmetric Gaussian.
  - $\mathbf{C}_{\mathrm{Tx}}$ and $\mathbf{C}_{\mathrm{Rx}}$ are transmit and receive covariance matrices.
- $\mathbf{X} = [\mathbf{x}_1, \ldots, \mathbf{x}_{N_P}] \in \mathbb{C}^{N_T \times N_P}$: denotes the unit modulus transmit QPSK signals, no power allocation/control.
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Signal Model

Conventional (sub-optimal) channel estimation schemes

- LS scheme:

$$\widehat{\mathbf{H}}_{LS} = \mathbf{Y}\mathbf{X}^H(\mathbf{X}\mathbf{X}^H)^{-1}$$

- MMSE scheme:

$$\widehat{\mathbf{H}}_{u-MMSE} = \mathbf{Y}\mathbf{X}^H(\mathbf{X}\mathbf{X}^H + \sigma^2\mathbf{I}_{N_R})^{-1}$$

Optimal channel estimation scheme for Gaussian channels

- MMSE scheme with knowledge of the spatial correlation matrices:

$$\operatorname{vec}\left(\widehat{\mathbf{H}}_{\mathrm{MMSE}}\right) = (\mathbf{C}_{\mathrm{Tx}}^T\mathbf{X}^* \otimes \mathbf{C}_{\mathrm{Rx}})\times \times \left[(\mathbf{X}^H\mathbf{C}_{\mathrm{Tx}}\mathbf{X})^T \otimes \mathbf{C}_{\mathrm{Rx}} + \sigma^2\mathbf{I}_{N_RN_P}\right]^{-1}\operatorname{vec}(\mathbf{Y})$$

where $\otimes$ denotes Kronecker product.

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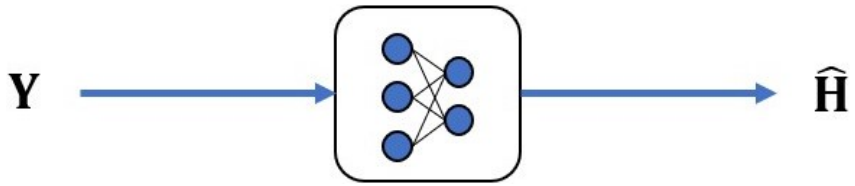
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Signal Model

Channel estimation can be modeled as a supervised learning problem.



Neural networks that can be used:

- Feed-forward Deep Neural Network (DNN)
- 1D Convolutional Neural Network (1D-CNN)
- 2D Convolutional Neural Network (2D-CNN)
- Graph Neural Network (GNN)

Channel Estimation as a *multi-output* regression problem

Input: Vector $\mathbf{t} = [\mathbf{y}_1, \ldots, \mathbf{y}_{N_P}]$ of *features* formed by stacking the real & imaginary parts of the received signal on transmitting N_P pilot symbols.

Output: Actual channel response $\mathbf{h} = [\mathbf{h}_{1,1}, \ldots, \mathbf{h}_{N_T,N_R}]$.

Training set:

- M examples of $\mathbf{t}_m$ stacked in $\mathbf{T} = \left[\mathbf{t}_1^T, \ldots, \mathbf{t}_M^T\right]^T \in \mathbb{R}^{M \times N_\mathrm{T} N_P}$
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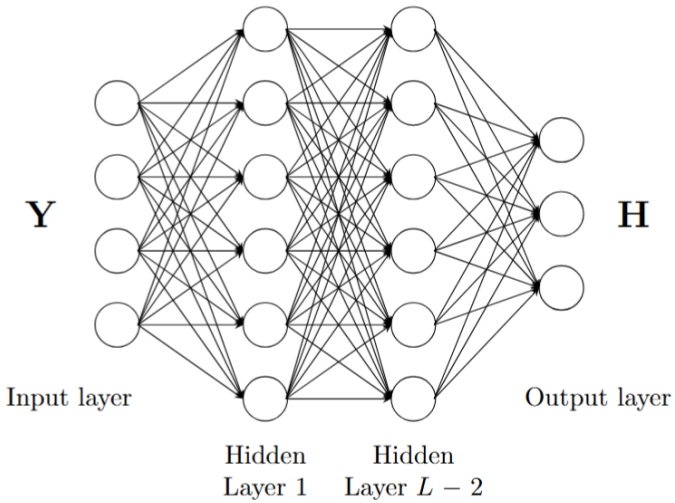
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- Multi-Layer Perceptron (MLP) regarded as universal function approximator.



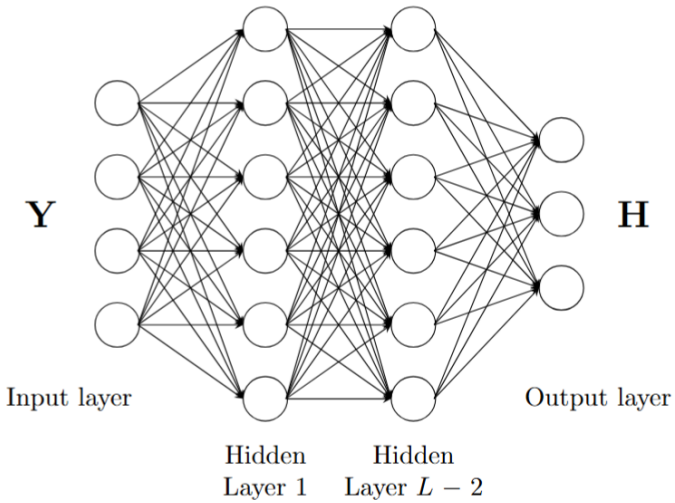
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$$\mathbf{r}_l = f_l\left(\mathbf{r}_{l-1}; \phi_l\right) = \sigma\left(\mathbf{W}_l \mathbf{r}_{l-1} + \mathbf{b}\right); \qquad l = 1 \ldots L$$

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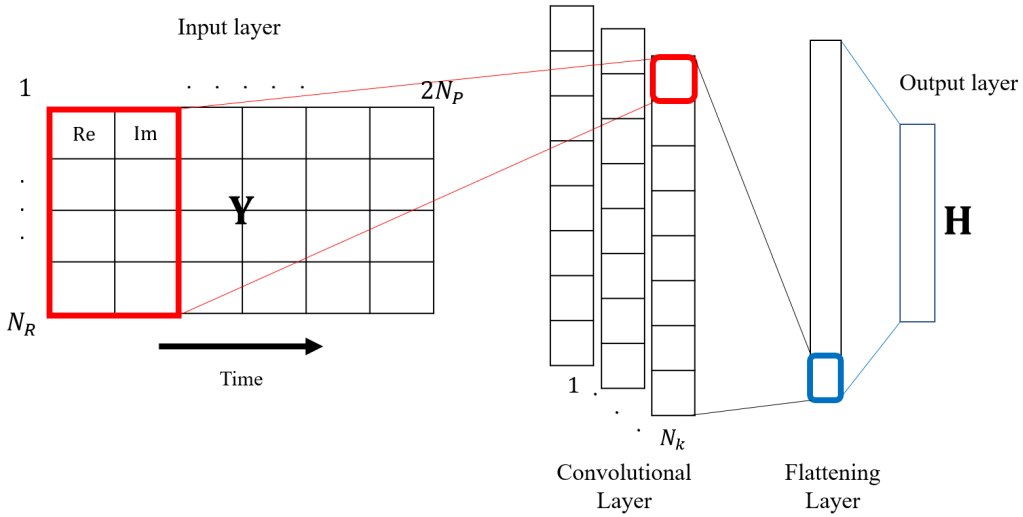
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- The kernels slide only in *one* direction. Suitable for processing time-series data.

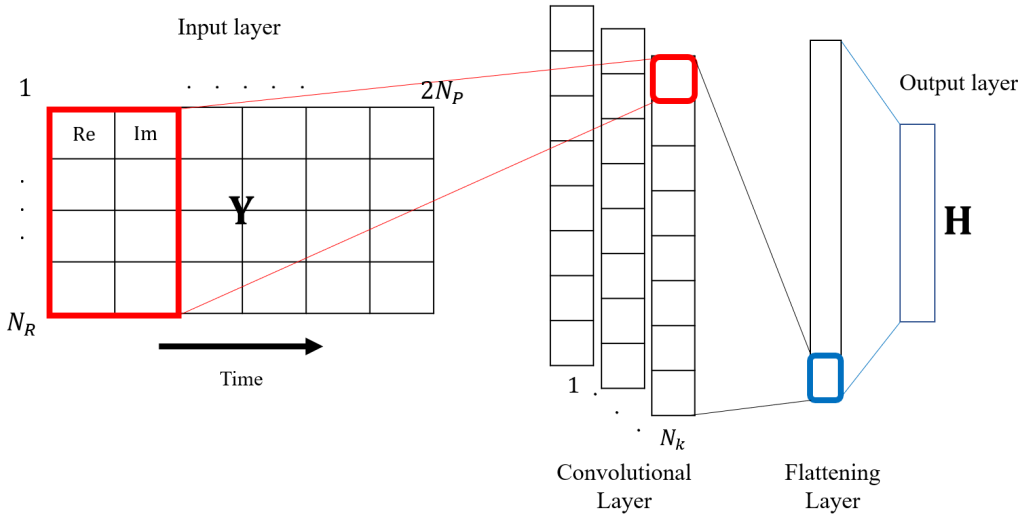


- 1 convolutional layer (shallow)
- Generated output feature map $\mathbf{Z}_l \in \mathbb{R}^{1\times(w_l-k_w+1)}$

$$(\mathbf{Z}_l)_n = \sum_{i=1}^{k_h}\sum_{j=1}^{k_w}(\mathbf{K}_l)_{i,j}(\mathbf{Y}_l)_{i,j+n-1}$$

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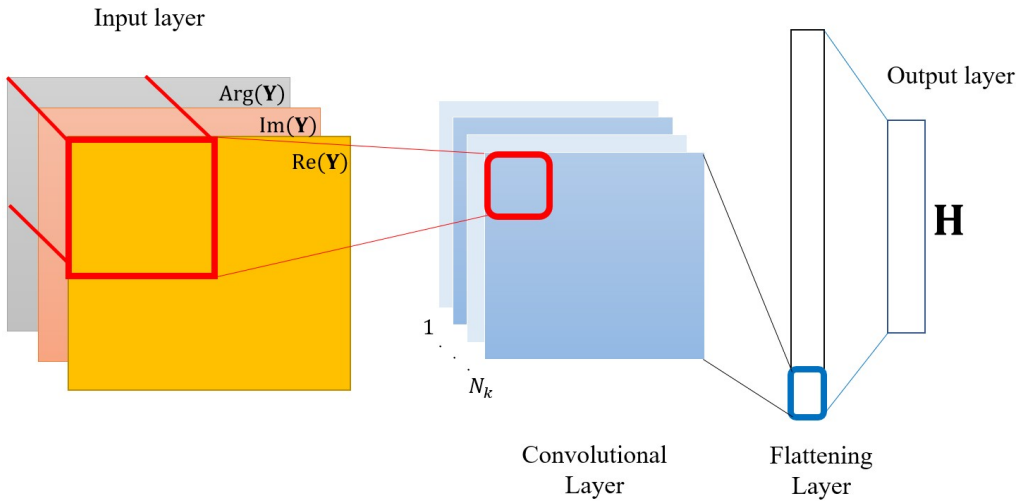


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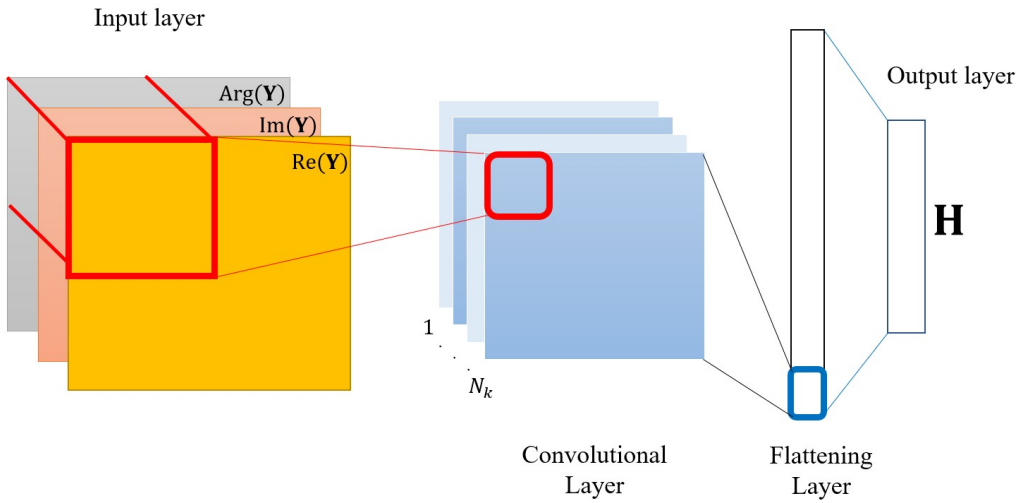


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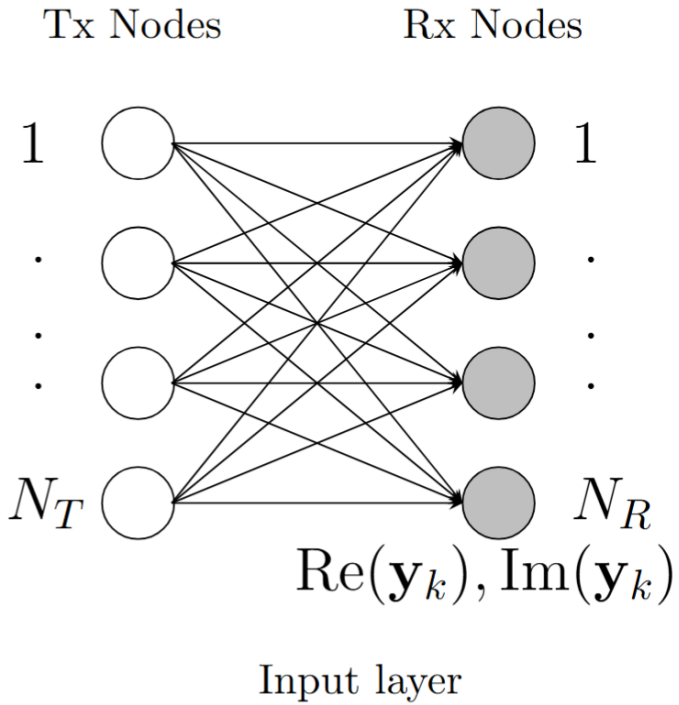


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NN Approach: GNN

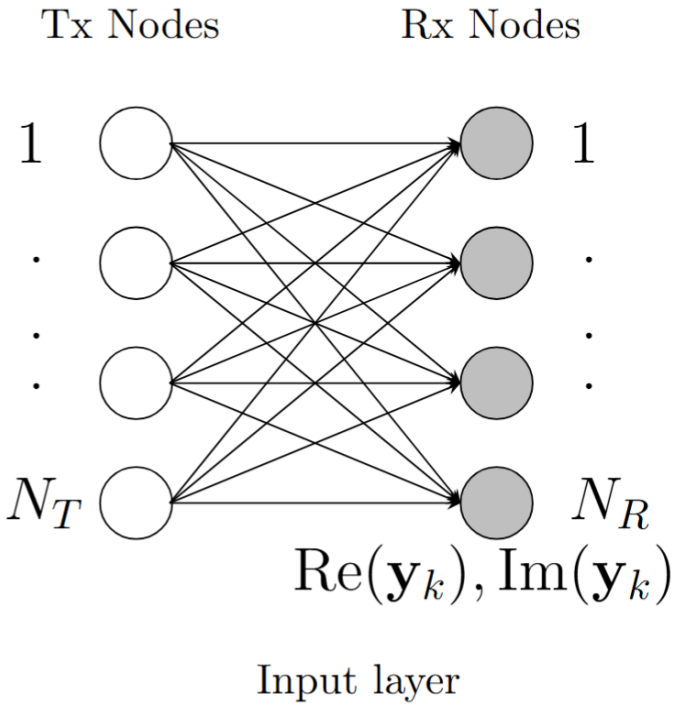
- The *edges* of graph represent the MIMO channel.



- Bipartite graph with N_T transmit and N_R receive nodes.
- Rx Nodes contain the received signals at each antenna i.e., $2N_P$ elements (real and imaginary parts of received pilots).

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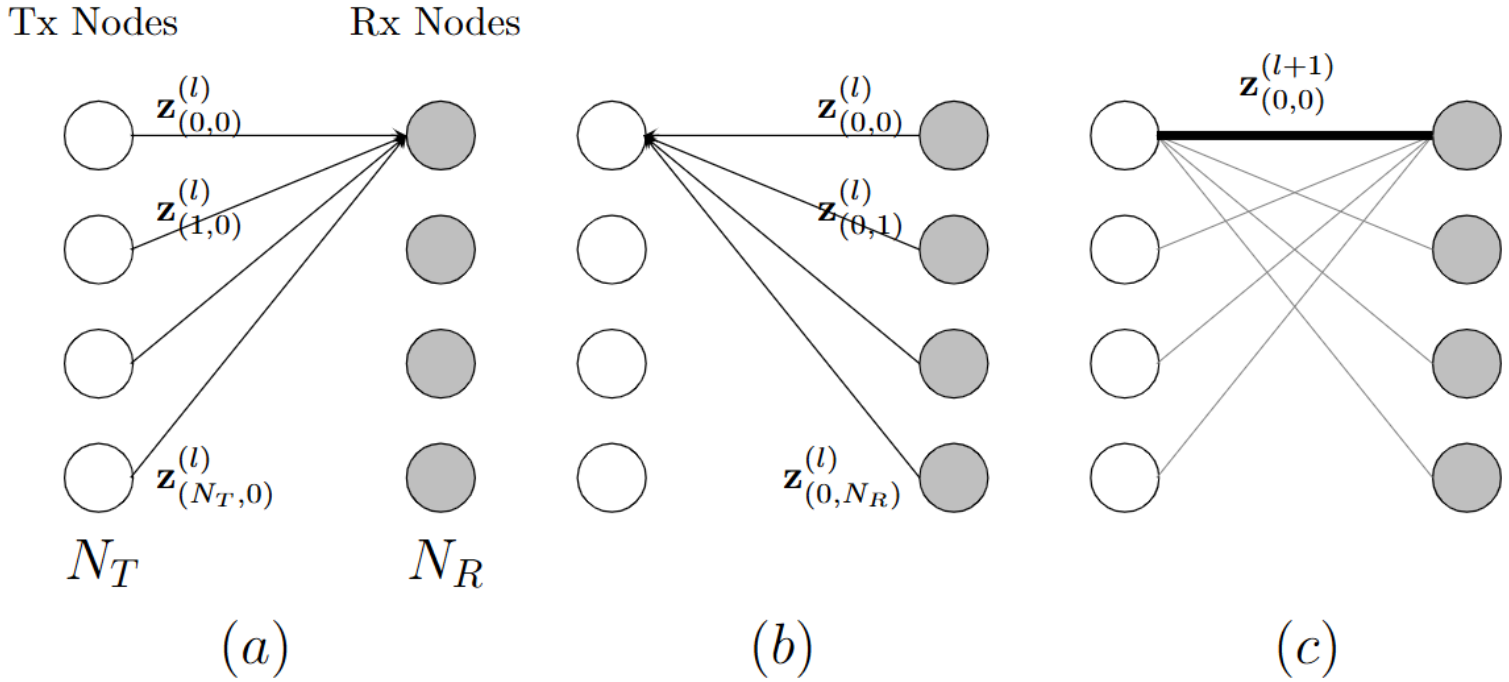
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NN Approach: GNN

- Edge-GNN makes use of Message Passing Algorithm (MPA) to learn the edge embedding over each hidden layer.

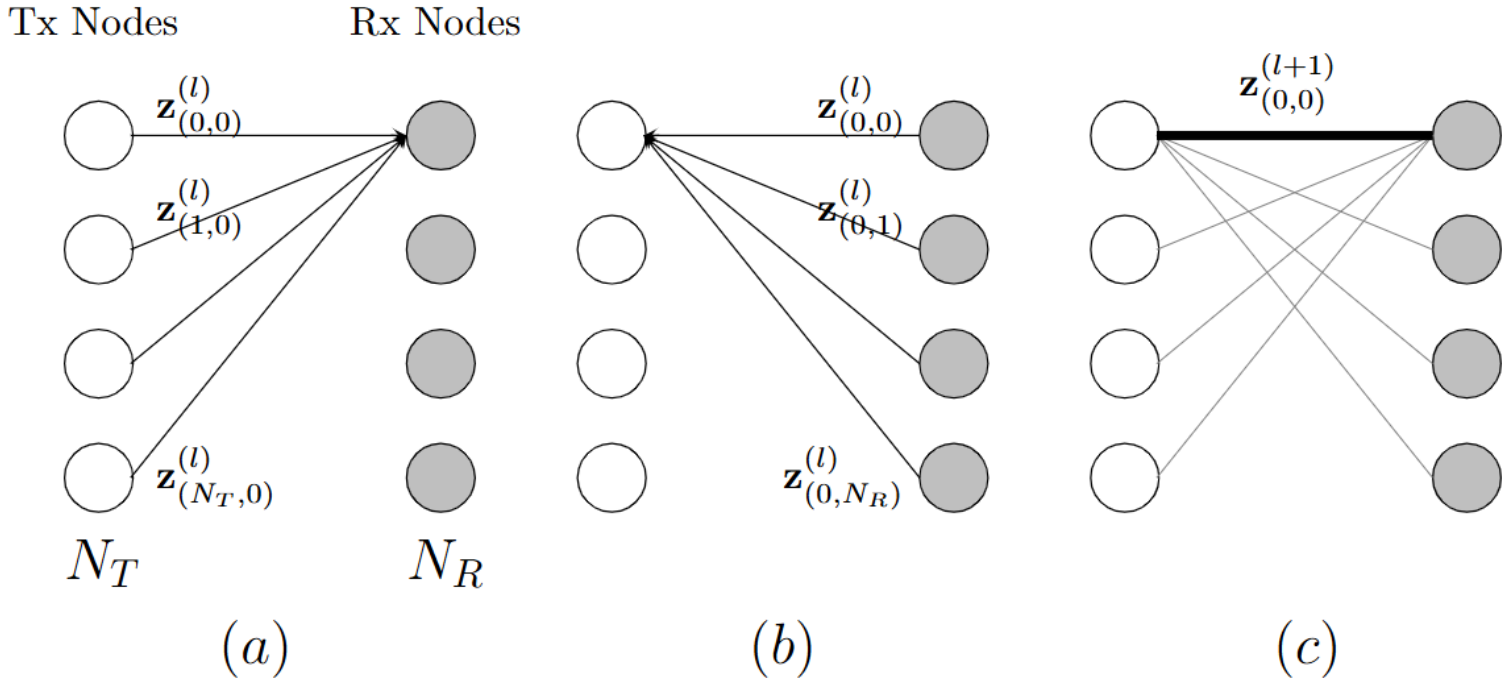


- Embedding of each edge (t, r) in layer l is denoted as $z^l_{(t,r)}$.
- MPA involves message aggregation and update rule.

$$\mathbf{z}^{(l+1)}_{(t,r)} = \sigma\left(W^{(l)}_{edge}\mathbf{z}^{(l)}_{(t,r)} + W^{(l)}_{t}\mathbf{m}^{(l)}_{\mathcal{N}(t)} + W^{(l)}_{r}\mathbf{m}^{(l)}_{\mathcal{N}(r)}\right)$$

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Parameter Set

Communication System:

- Scenarios: $N_{\mathrm{T}} = N_{\mathrm{R}} = 2, 4, 8, 16$ transmit/receive antennas.
- Highly correlated MIMO channel: correlation matrices $\mathbf{C}_{\mathrm{Tx}}$ and $\mathbf{C}_{\mathrm{Rx}}$ are chosen as Toeplitz matrices with n-th diagonal fixed to $0.9^{|n|}$
- Performance measure: Normalised Mean Squared Error (NMSE).

Machine Learning schemes:

- Training set: $M = 10^4$ and $N_P = N_{\mathrm{T}}$ fixed pilot symbols per channel. Test set: 10^3 channels.
- Benchmarks: Conventional estimation (LS, MMSE), *shallow* 1D-CNN architectures.

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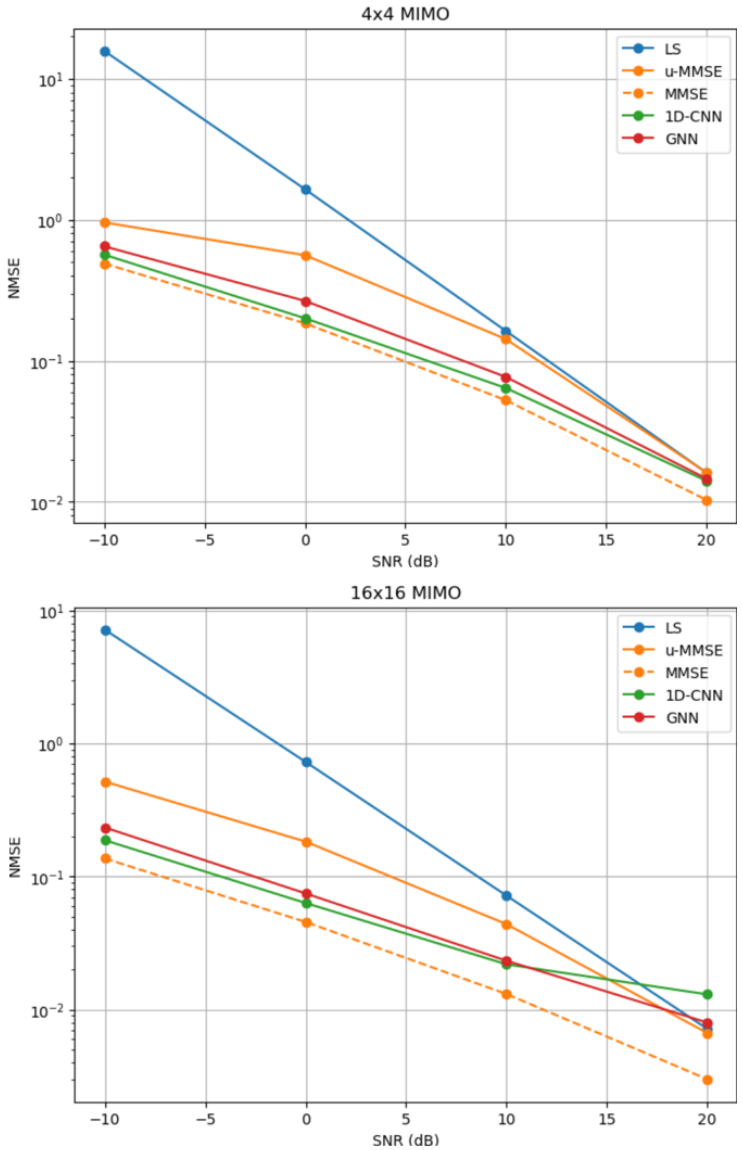
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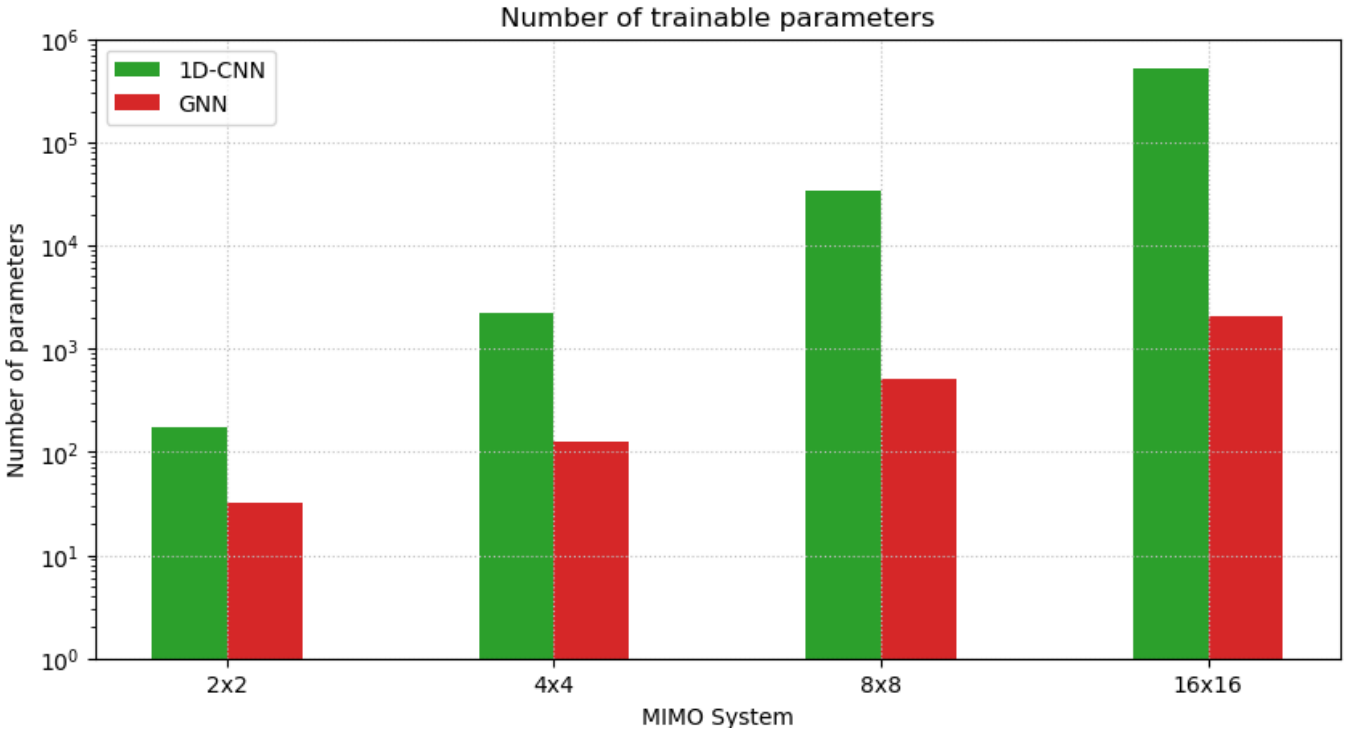
Estimation Error NMSE vs. SNR



- Benchmarks: LS, u-MMSE, MMSE, and shallow 1D-CNN architecture.
- GNN architectures outperform LS/u-MMSE: GNN learns the channel statistics.
  - At least $+1$ order of magnitude compared to LS at low SNR.
- GNN offers similar performance to 1D-CNN with lower computational complexity.
- For 16×16 MIMO, GNN performs better than 1D-CNN at 20dB.

Computational Complexity Analysis (Training Phase)

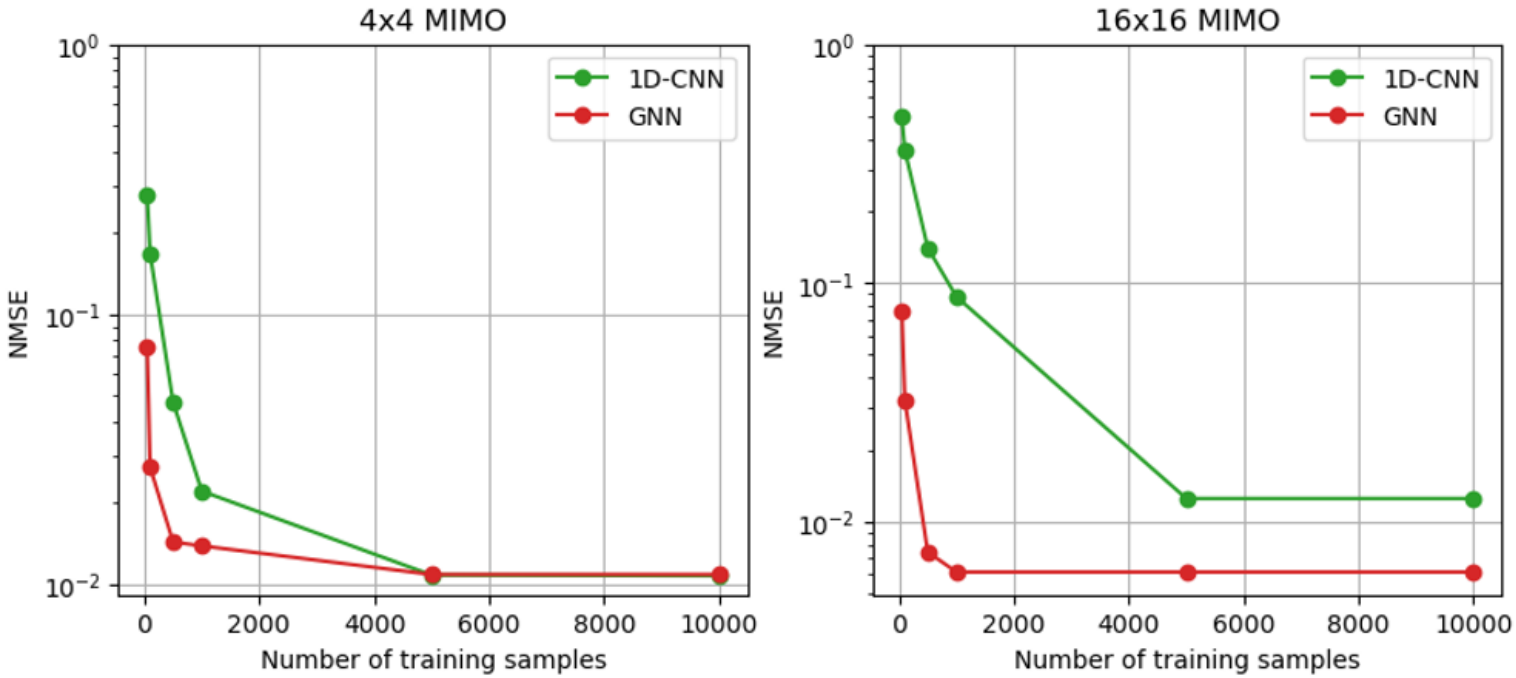
Total number of trainable parameters



- GNNs employ lesser number of parameters than 1D-CNNs.
- The difference in the number of trainable parameters increases as the MIMO system scales up.

Computational Complexity Analysis (Training Phase)

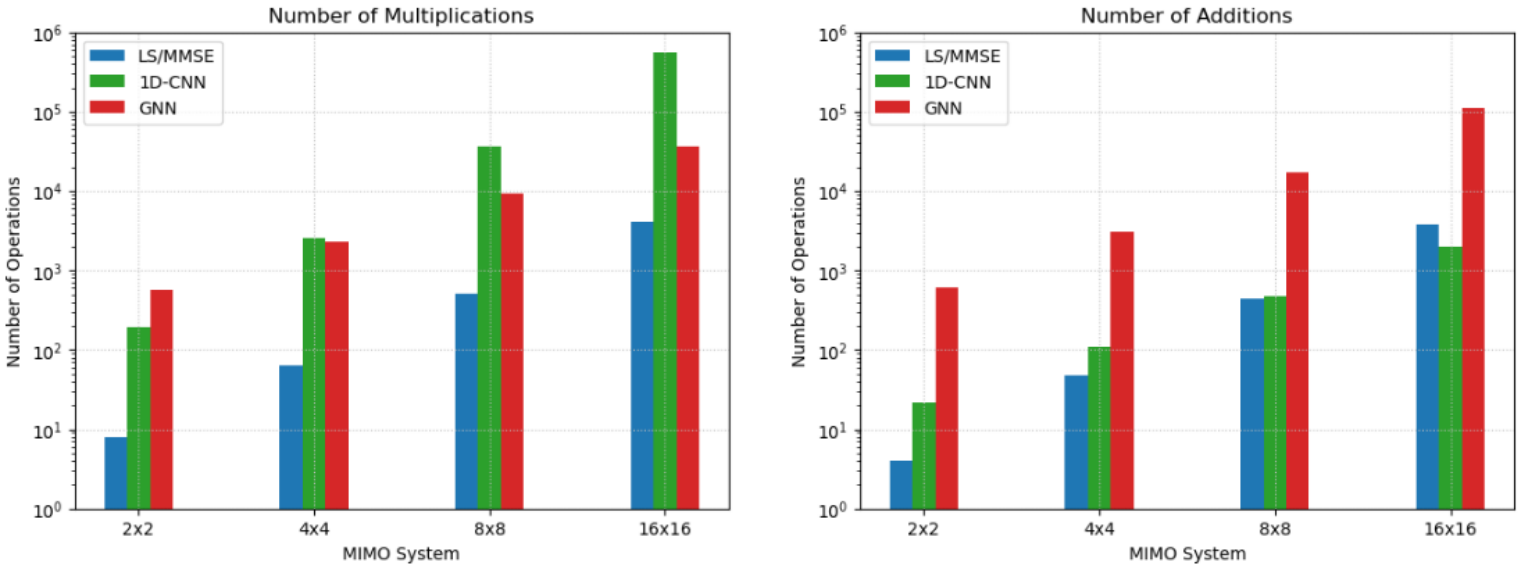
Training Dataset Size



- GNN architectures reach convergence faster than 1D-CNNs.
- For 16×16 MIMO, 1D-CNNs demand a larger training dataset to reach convergence.

Computational Complexity Analysis (Prediction Phase)

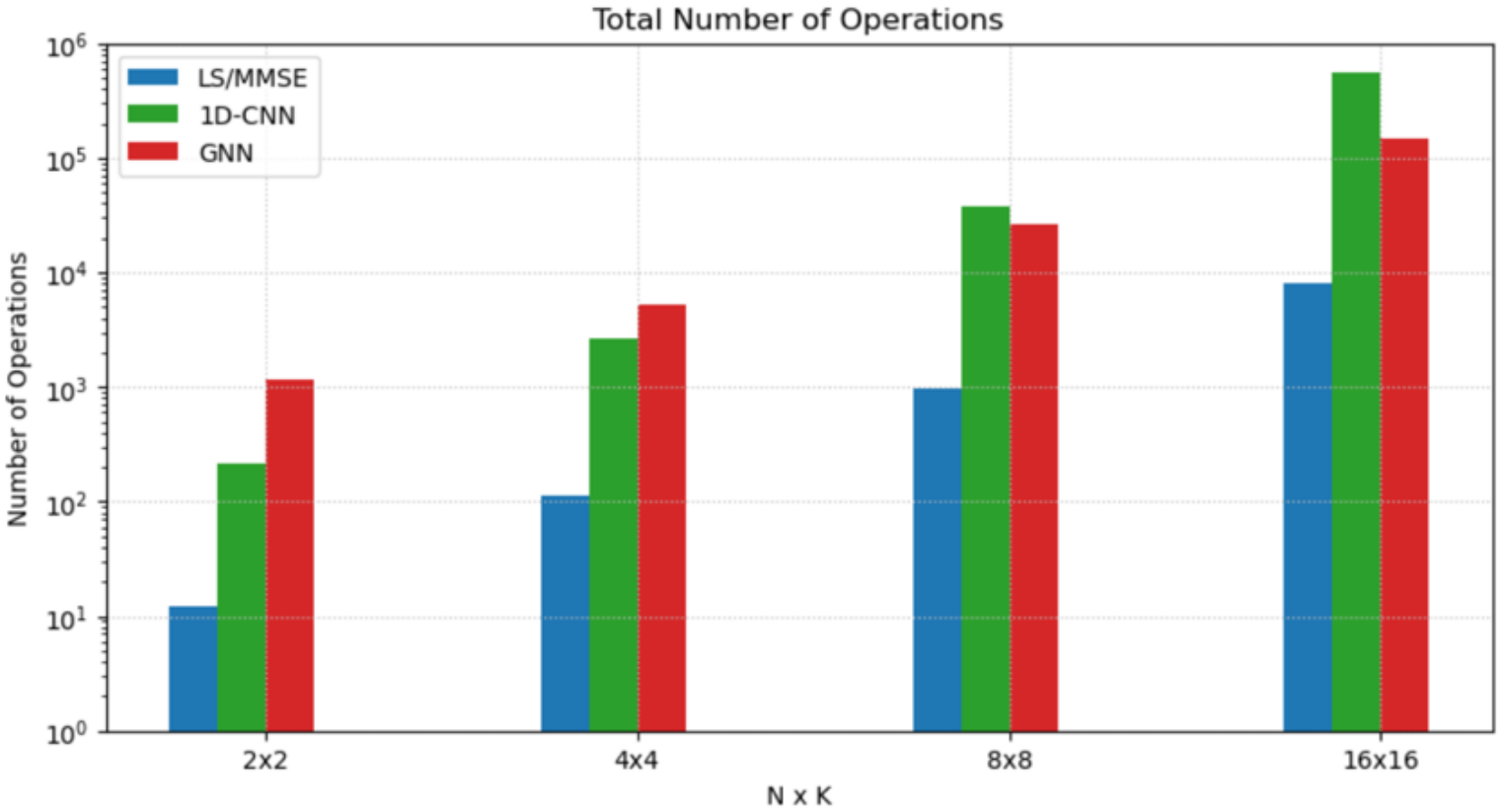
Number of Operations



- GNN-based estimators require at least $10\times$ more multiplication and $100\times$ more addition operations than those of LS/u-MMSE estimators.
- GNNs record a higher number of additions than 1D-CNNs (owing to the message passing algorithm).

Computational Complexity Analysis (Prediction Phase)

Total Number of Operations



- At higher MIMO settings, GNNs require *lesser* number of total operations than 1D-CNN models.

Conclusions

We have investigated the effect of using different neural network architectures for MIMO channel estimation.

- 1D-CNN and GNN outperform linear conventional schemes.
- Performance-Complexity tradeoff: GNN is the better choice than shallow 1D-CNN.
  - Employs lower number of trainable parameters than 1D-CNN.
  - Rate of convergence to a given dataset is faster.
  - For large MIMO settings, GNN records lower total number of operations (additions/multiplications) than 1D-CNN.

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Publications

Conferences

(C1) **D. R. Kumar**, C. Antón-Haro and X. Mestre, "Graph Neural Network Architecture for MIMO Channel Estimation," 2025 IEEE Consumer Communications and Networking Conference, Las Vegas, USA, 2025.

(C2) **D. R. Kumar**, C. Antón-Haro and X. Mestre, "Shallow Neural Networks for Channel Estimation in Multi-Antenna Systems," IEEE INFOCOM 2023 - IEEE Conference on Computer Communications Workshops (INFOCOM WKSHPS), Hoboken, NJ, USA, 2023, pp. 1-6, doi: 10.1109/INFOCOMWKSHPS57453.2023.10225983.

Other Publications

(O1) **D. R. Kumar**, C. Antón-Haro, and X. Mestre, "Performance-Complexity Tradeoff Analysis for MIMO Channel Estimation: GNN vs. CNN Architectures," Proceedings of 8th Technical Meeting of INTERACT COST Action, Helsinki, Finland, 2024.

(O2) **D. R. Kumar**, C. Antón-Haro, and X. Mestre, "Neural Network-based MIMO Channel Estimation: Shallow vs. Deep Architectures," Proceedings of 4th Technical Meeting of INTERACT COST Action, Dubrovnik, Croatia, 2023.

(O3) **D. R. Kumar**, C. Antón-Haro, and X. Mestre, "Deep Learning-based Channel Estimation and Data Detection for Multi-Antenna Systems," Proceedings of 2nd Technical Meeting of INTERACT COST Action, Lyon, France, 2022.

Outline

Introduction

Part I - Channel Estimation

Part II - Symbol Detection

Part III - Precoder Optimization

Conclusions

Motivation

- Traditional Orthogonal Multiple Access (OMA) schemes versus Non-orthogonal Multiple Access (NOMA) schemes.
  - Superposition coding at the transmitters.
  - Successive Interference Cancellation (SIC) at the receivers.
- Challenges: SIC operation subject to error propagation from channel estimation errors, synchronization issues, etc.
- Non-linear channels: impact of signal clipping.
- Machine learning approaches suitable for overcoming such limitations.

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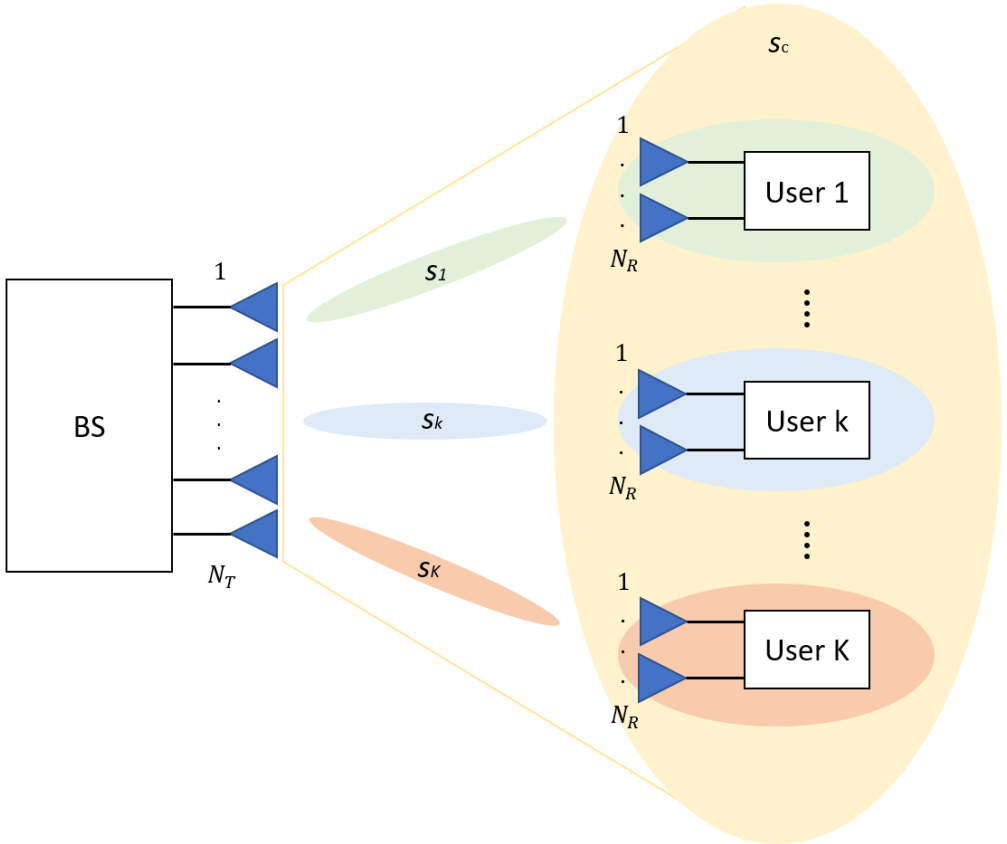
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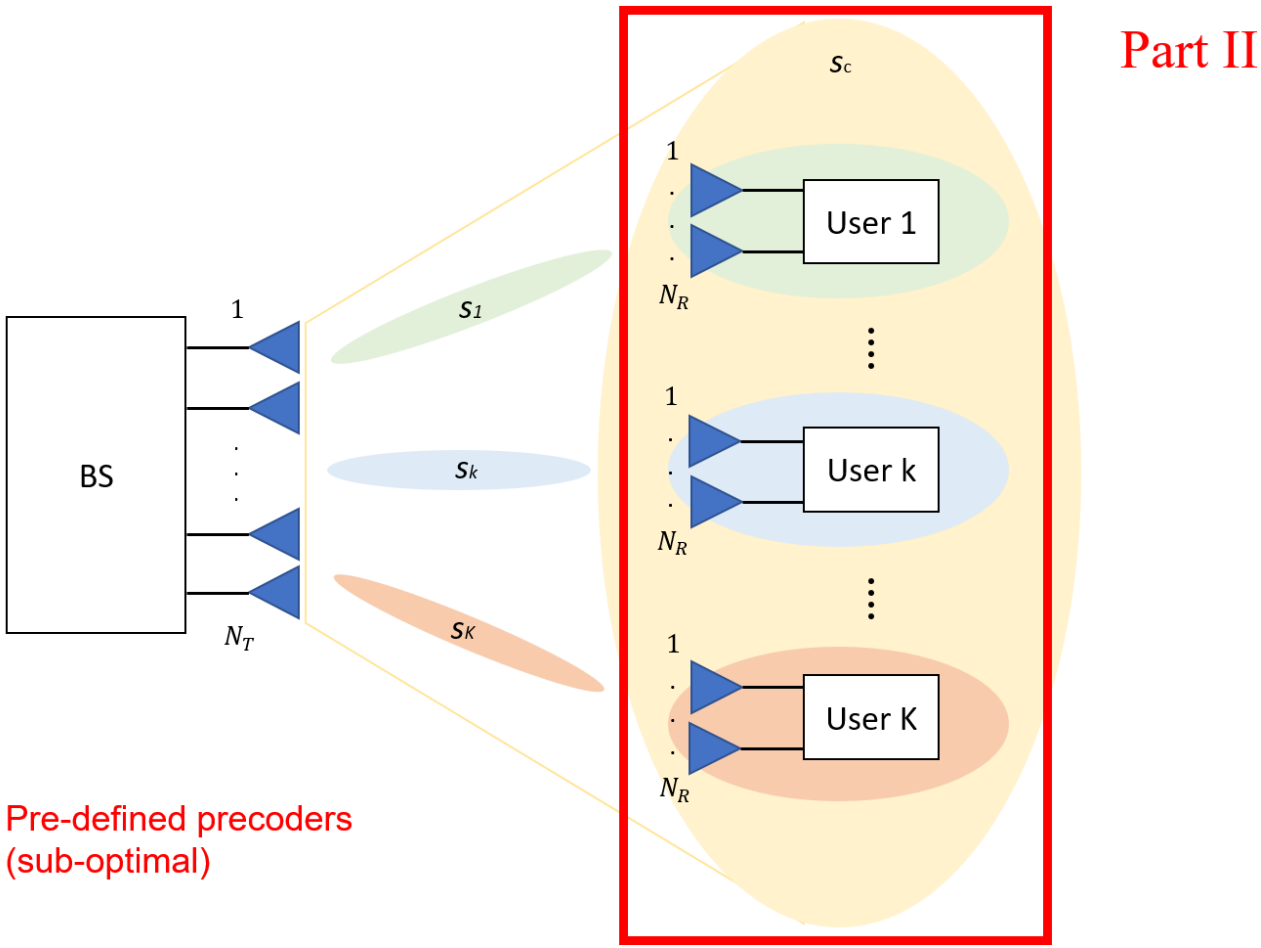
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1-layer RSMA

- Rate Splitting Multiple Access (RSMA): common stream broadcasted to all users + private stream for each user.



1-layer RSMA


Part II
s_c
1
N_R
User 1
s_1
BS
1
N_T
s_k
1
N_R
User k
s_K
1
N_R
User K
Pre-defined precoders
(sub-optimal)

Signal Model

Transmit signal in a MIMO-RSMA setting:

$$\mathbf{x} = \alpha_c \mathbf{p}_c s_c + \sum_{k=1}^{K} \alpha_k \mathbf{p}_k s_k$$

Received signal at user k with linear channels:

$$\mathbf{y}_k = \mathbf{H}_k \alpha_c \mathbf{p}_c s_c + \mathbf{H}_k \alpha_k \mathbf{p}_k s_k + \sum_{j \neq k} \mathbf{H}_k \alpha_j \mathbf{p}_j s_j + \mathbf{n}_k$$

- $\mathbf{P} = [\mathbf{p}_c, \mathbf{p}_1, \ldots, \mathbf{p}_k]^T \in \mathbb{C}^{N_T \times K+1}$ with $\mathbf{p}_c, \mathbf{p}_k \in \mathbb{C}^{N_T \times 1}$ precoding matrix. ZFBF for the private streams
- $\alpha_c, \alpha_k \in [0, 1]$: power allocation; such that $\alpha_c^2 + \sum_{k=1}^{K} \alpha_k^2 = 1$
- $\mathbf{H}_k \in \mathbb{C}^{N_R \times N_T}$: channel matrix of kth user
- $\mathbf{n}_k$: i.i.d. AWGN

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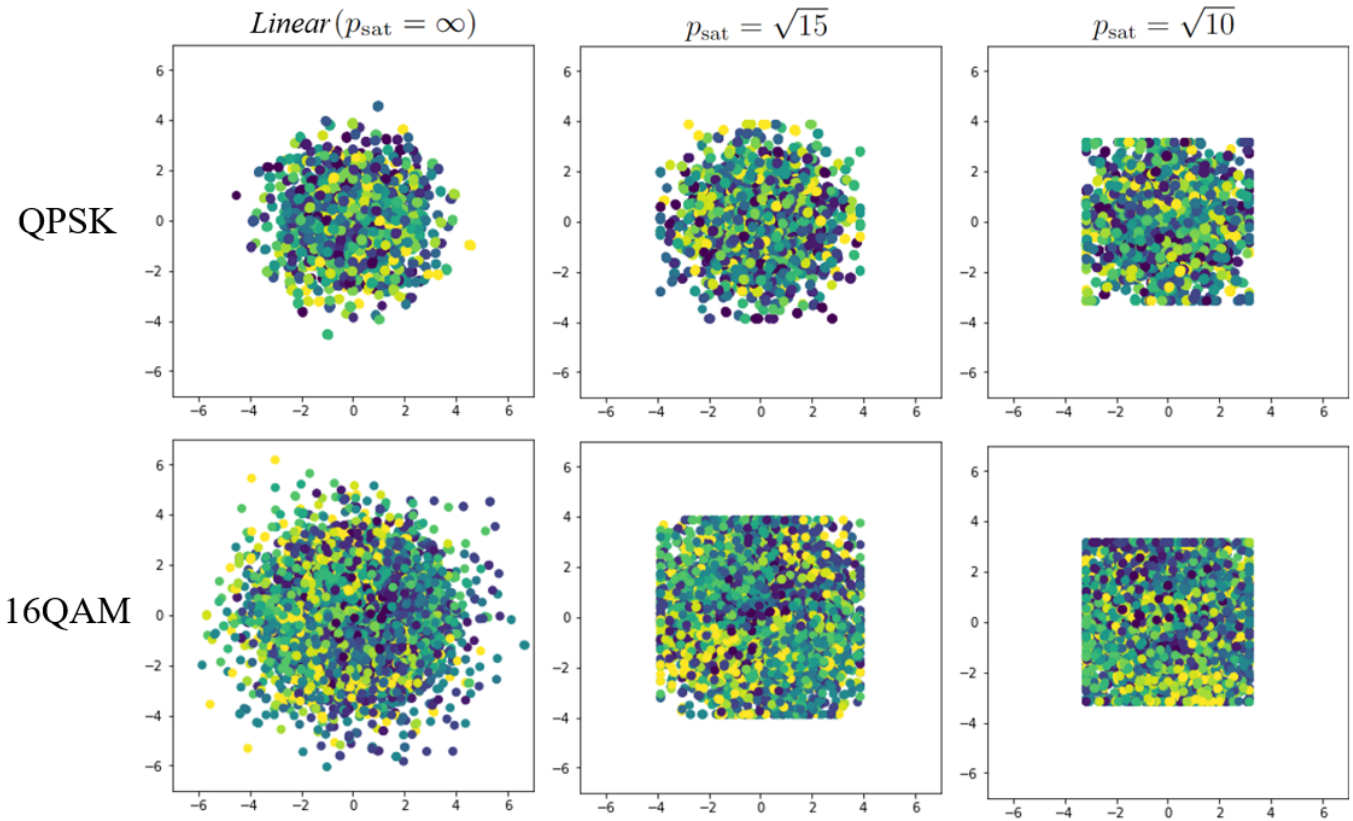
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Signal Model

Non-linear channels: signal clipping at the receiver amplifiers.

$$\phi_{\text{AM-AM}}(x) = \begin{cases} x & \text{for } x \leq p_{\text{sat}} \\ p_{\text{sat}} & \text{for } x > p_{\text{sat}} \end{cases}$$



Problem Formulation

Goal

To design the Deep Neural Network (DNN) equivalents for performing symbol detection and SIC operation in 1-layer MIMO-RSMA setting.

- Symbol detection can be cast into a *multi-class* classification task.
- Performance in linear and non-linear channels.

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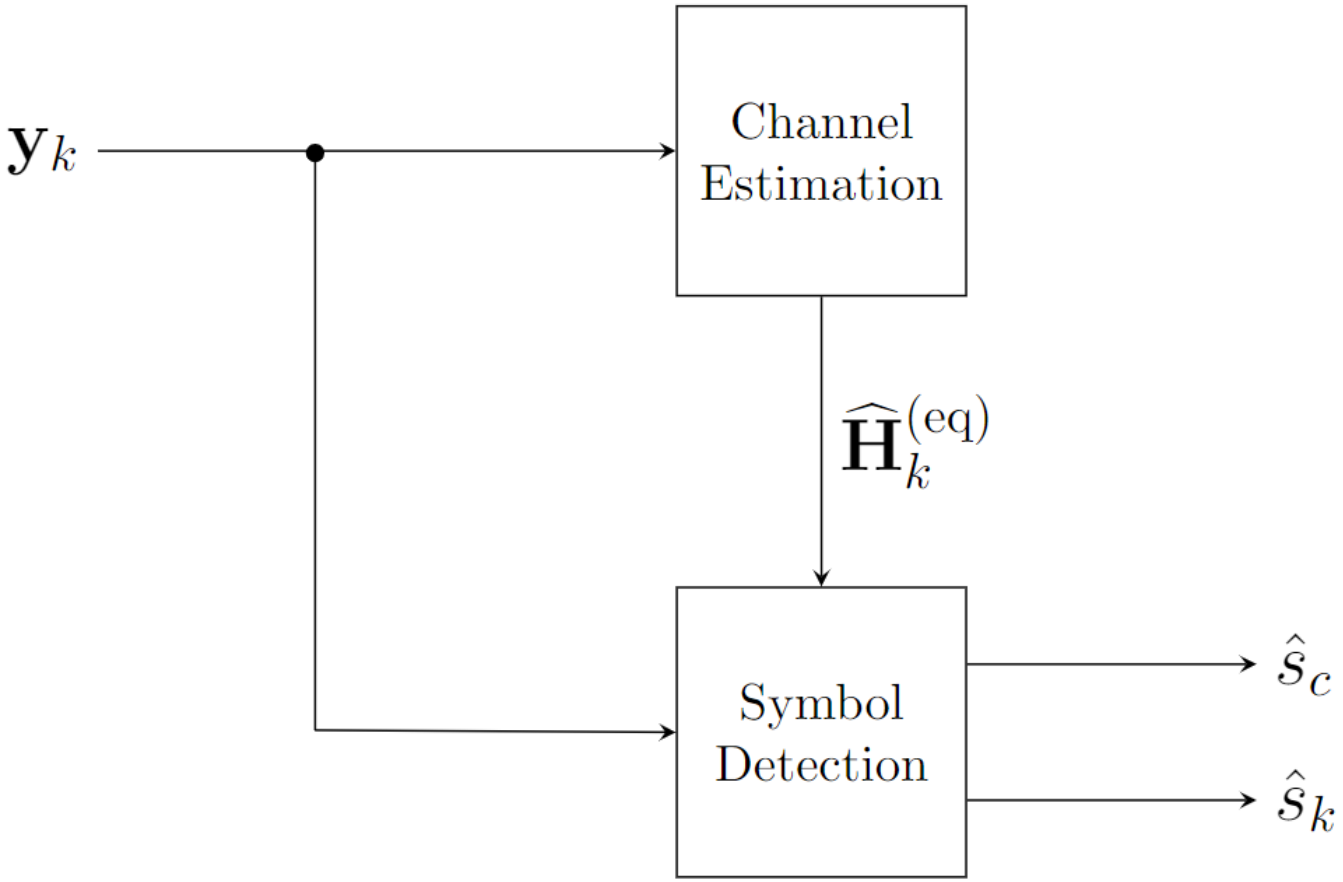
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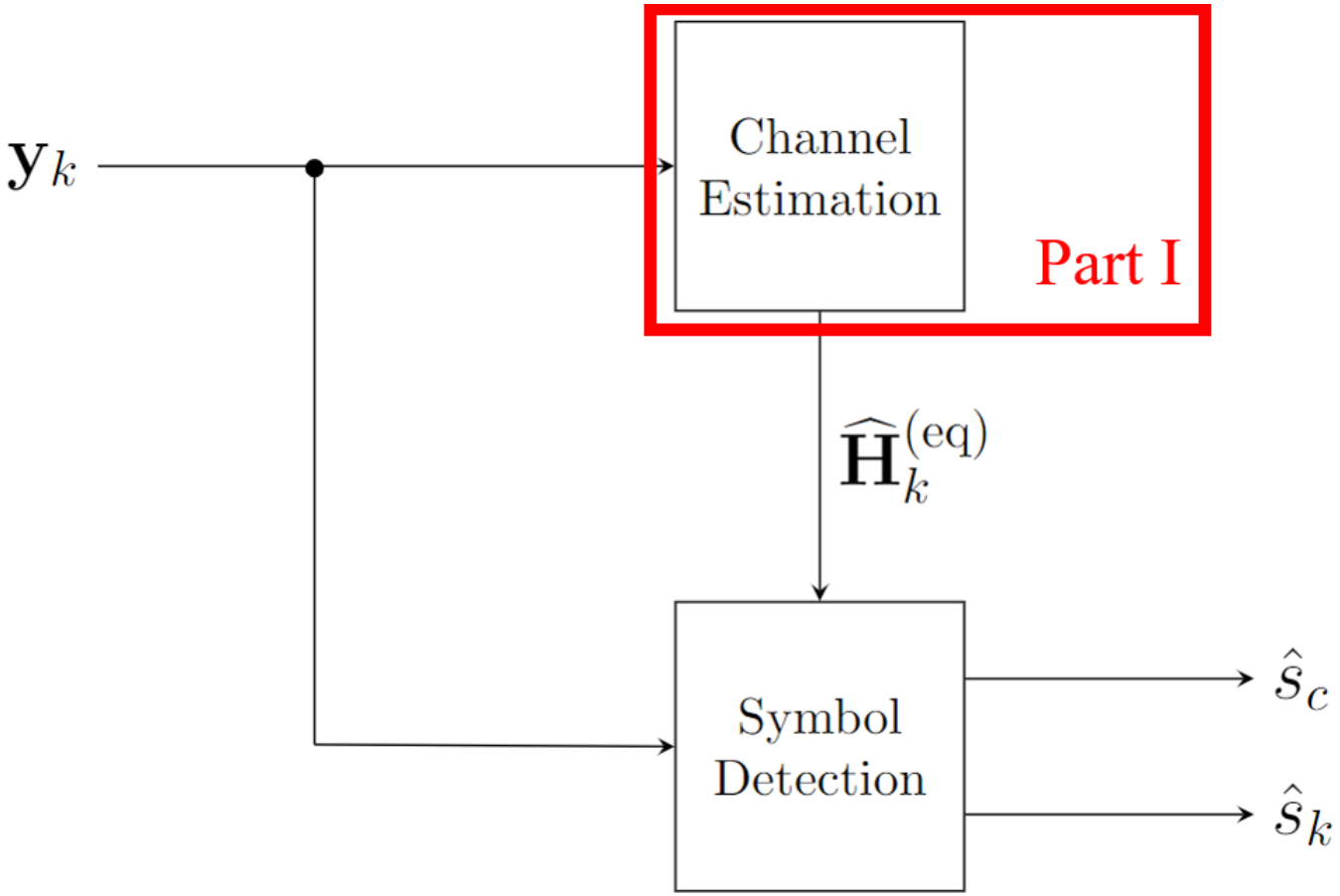
Joint-Net scheme

- Joint detection of RSMA signals via black-box approach.



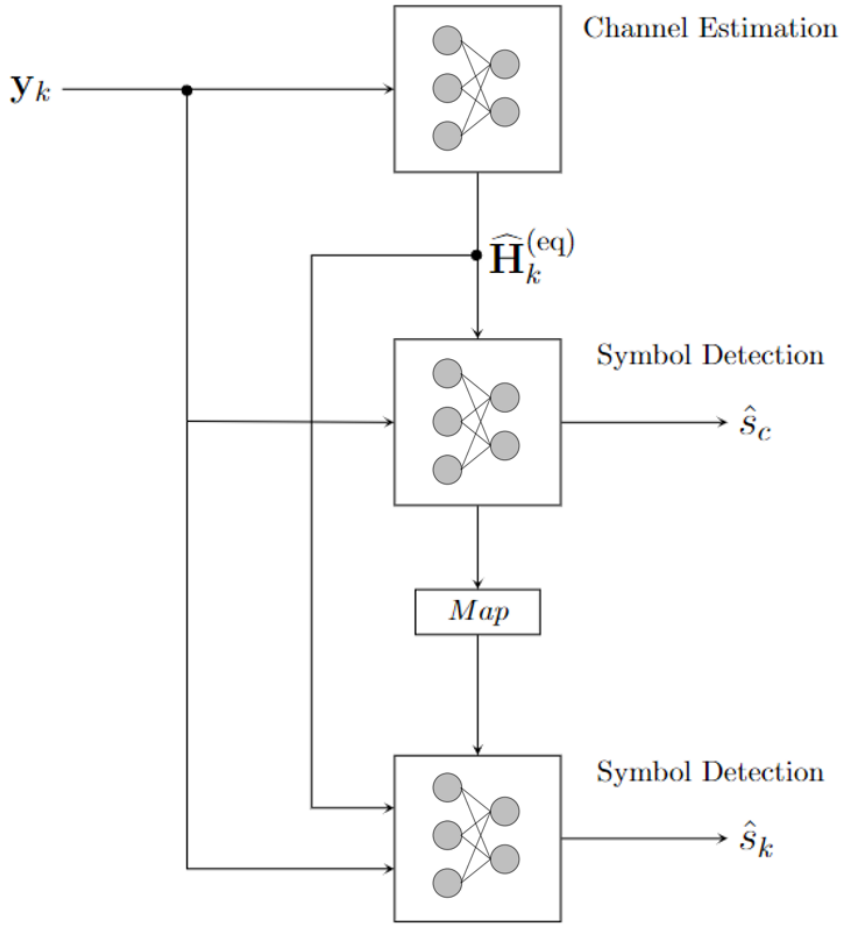
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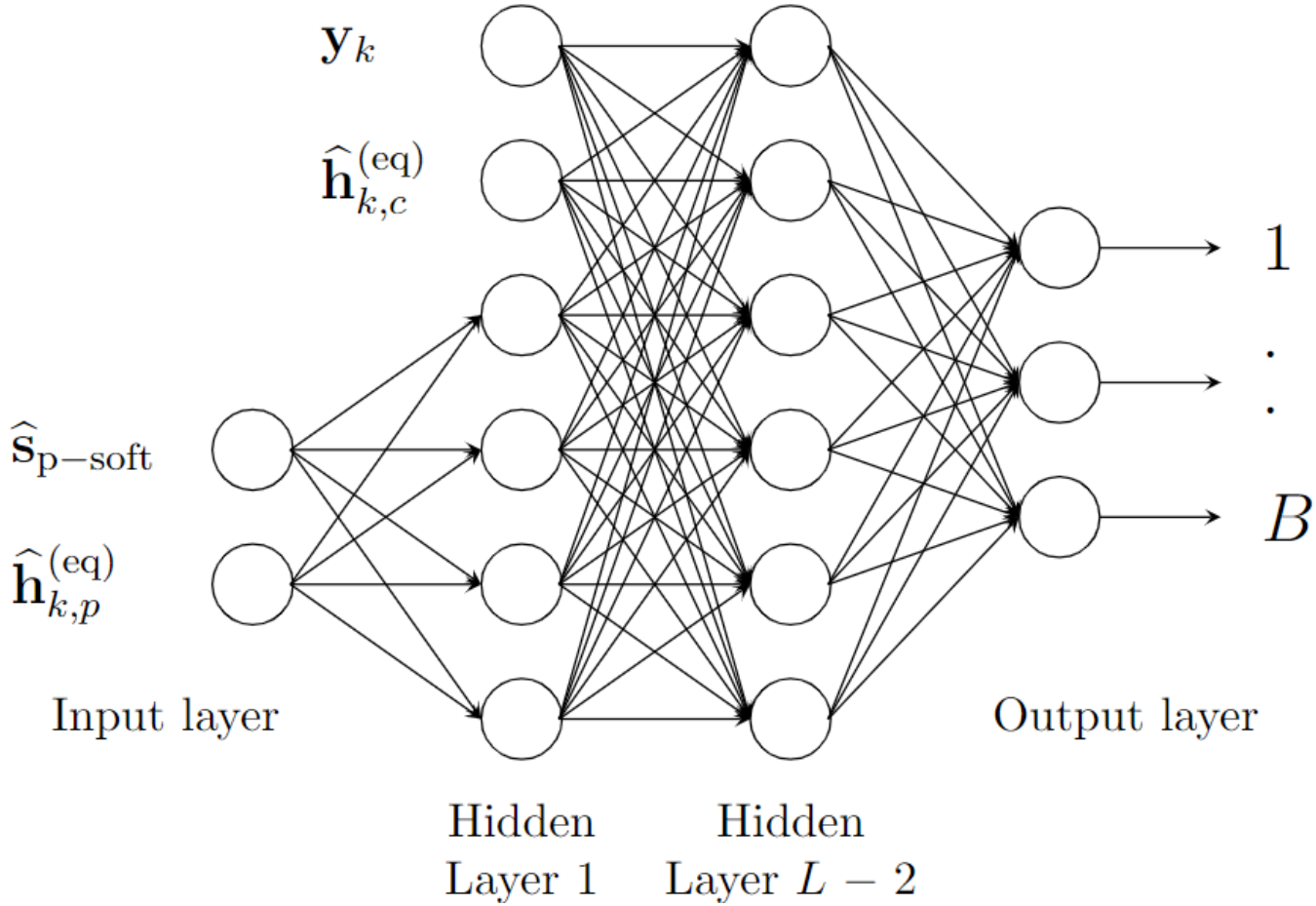
RS-Net scheme

Symbol (data) detection and channel estimation blocks replaced by DNN equivalents. Resemblance to SIC approaches.



RS-Net+ scheme

- Neural Network-enhanced receivers for MIMO-RSMA.



Soft estimate of private message symbol ($\hat{s}_{\mathrm{p-soft,k}}$)

- In multi-antenna users: to take advantage of the additional spatial degree of freedom.
- Intuition: To facilitate the trained common message detection to implicitly reconstruct and subtract the contribution of the private message from the received signal.
- Computation for the k-th user as:

$$\hat{s}_{\mathrm{p-soft,k}} = \frac{(\mathbf{H}_k \alpha_k \mathbf{p}_k)^H \mathbf{P}_c^{\perp} \mathbf{y}_k}{\sigma^2 + (\mathbf{H}_k \alpha_k \mathbf{p}_k)^H \mathbf{P}_c^{\perp} \mathbf{h}_k \alpha_k \mathbf{p}_k}$$

where $\mathbf{P}_c^{\perp}$ is the orthogonal projection matrix (which is only defined for $N_R > 1$) onto the null space of the equivalent common message channel vector $\mathbf{H}_k \mathbf{p}_c$, that is

$$\mathbf{P}_c^{\perp} = \mathbf{I}_{N_R} - \frac{\mathbf{H}_k \mathbf{p}_c \mathbf{p}_c^H \mathbf{H}_k^H}{\mathbf{p}_c^H \mathbf{H}_k^H \mathbf{H}_k \mathbf{p}_c}$$

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Parameter set (unless otherwise stated)

Communication System:

- $K = 2$ *single-* and *multi-* antenna users; $N_{\mathrm{T}} = \{3, 6\}$ transmit antennas, respectively.
- MIMO channel: AWGN model, with the entries being i.i.d. circularly symmetric Gaussian-distributed with zero mean and unit variance.
- Non-linear channels: power saturation $p_{\mathrm{sat}} = \{\sqrt{10}, \sqrt{15}\}$ for QPSK and 16QAM, respectively.
- Performance measures: Symbol Error Rate (SER) for symbol detection.

Machine Learning schemes:

- Channel estimation, $N_{\mathrm{ch}} = 10,000$ and $N_{\mathrm{P}} = 3$ fixed pilot symbols per channel. 20% used for validation.
- Symbol detection, $N_{\mathrm{ch}} = 10,000$ and $N_{\mathrm{D}} = 10$ random data symbols per channel.

Parameter set (unless otherwise stated)

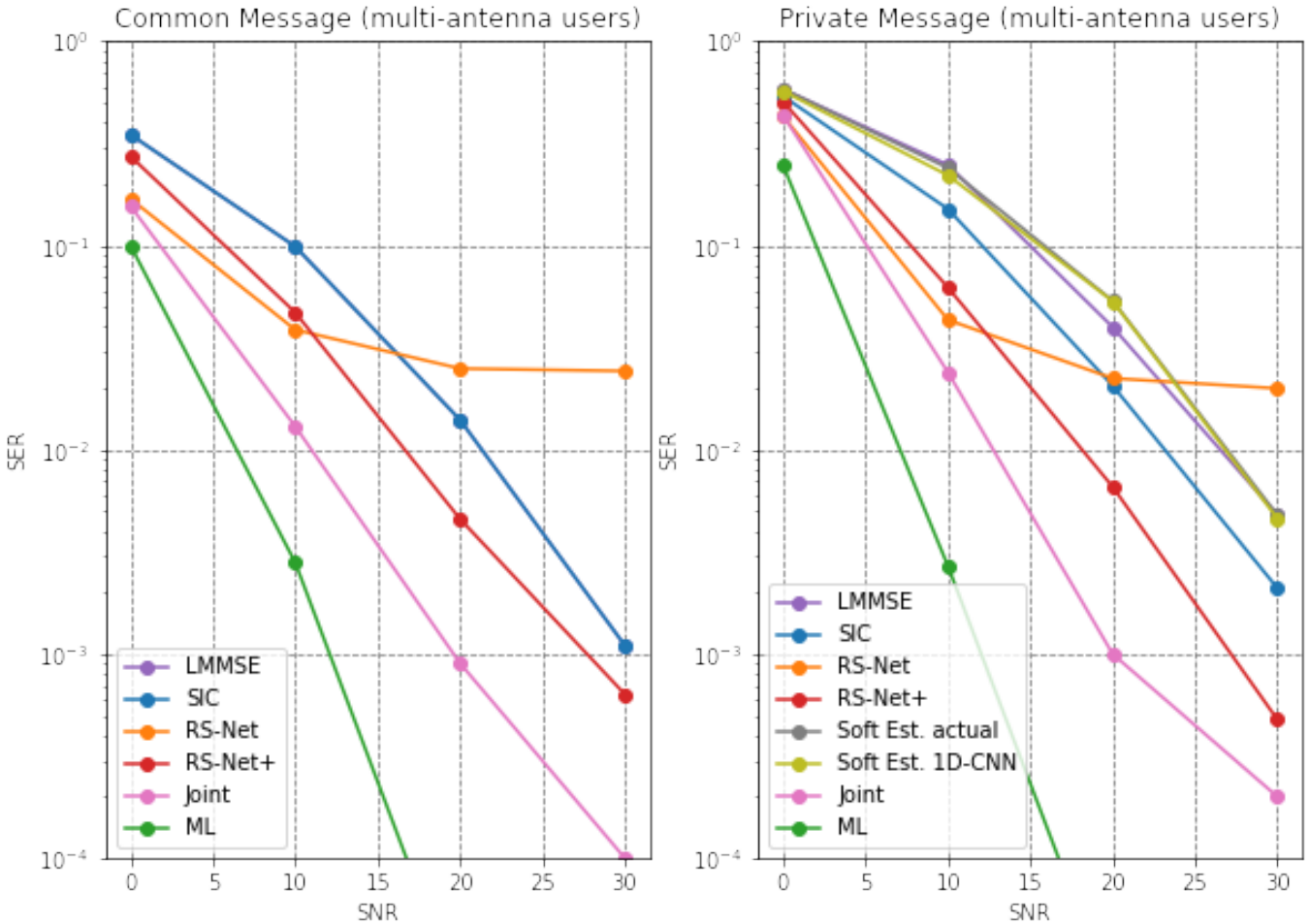
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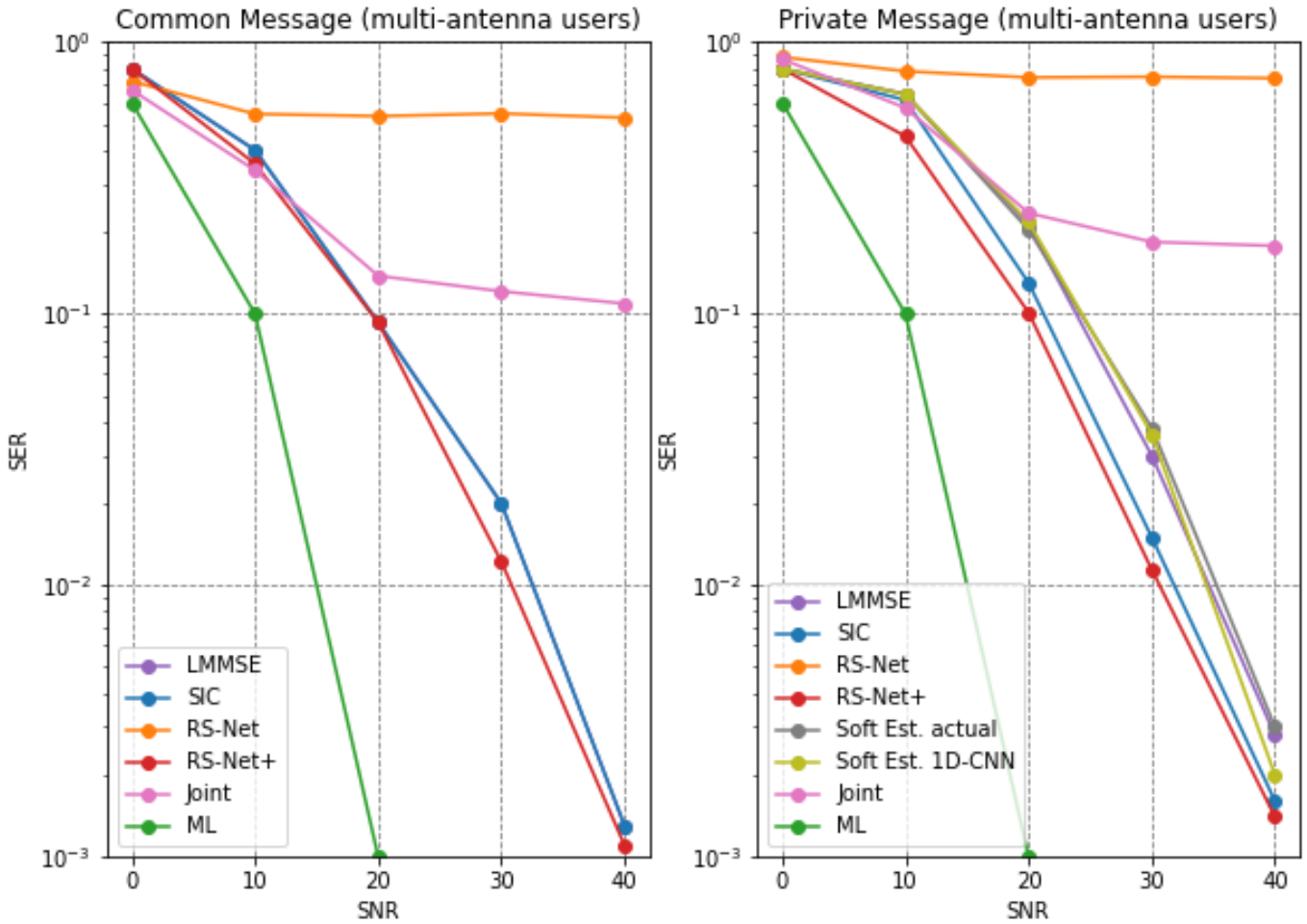
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Linear channels (multi-antenna users, QPSK)



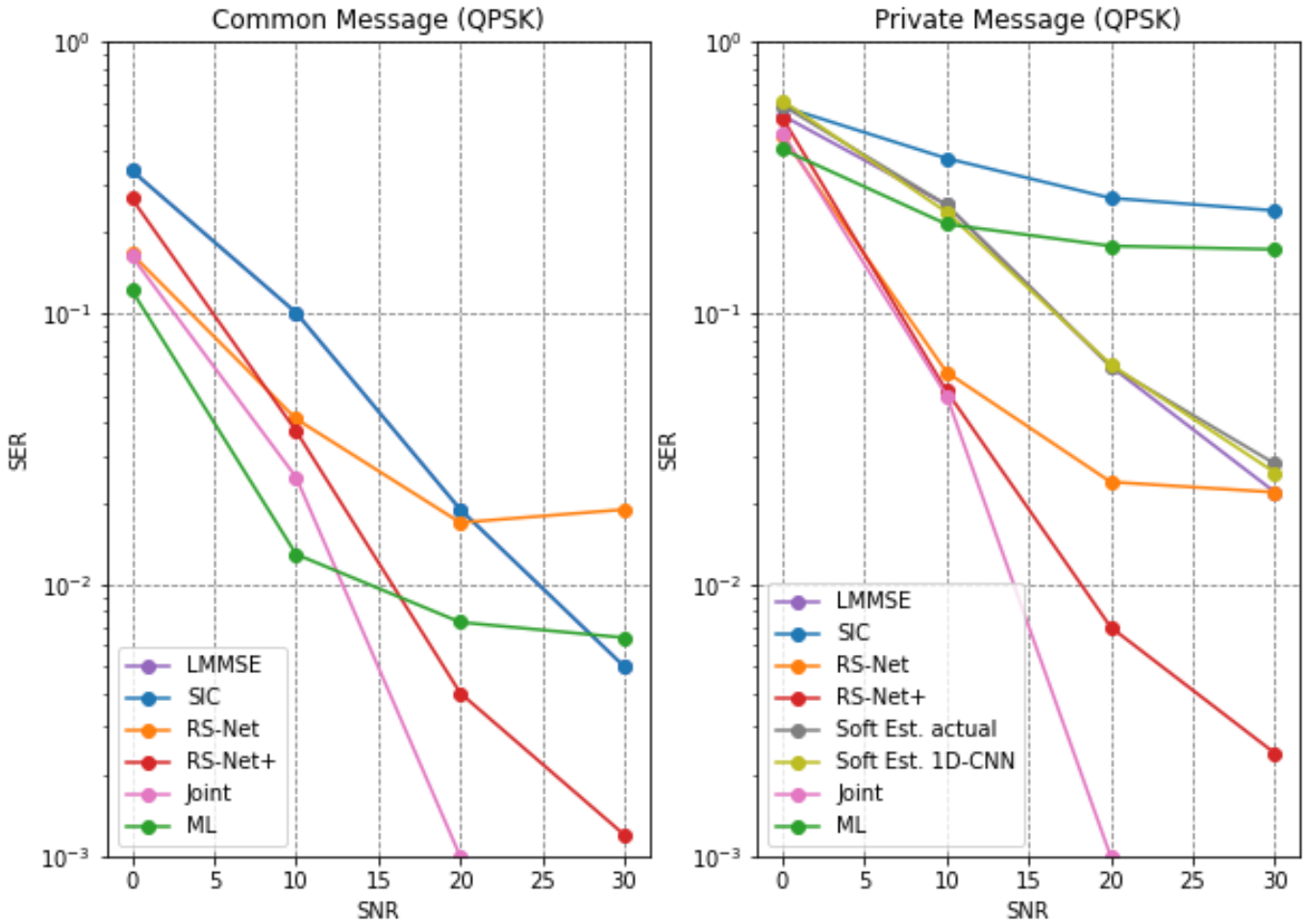
- RS-Net exhibits error floor for both common and private SER.
- RS-Net+ and Joint-Net outperform RS-Net and SIC.

Linear channels (multi-antenna users, 16QAM)



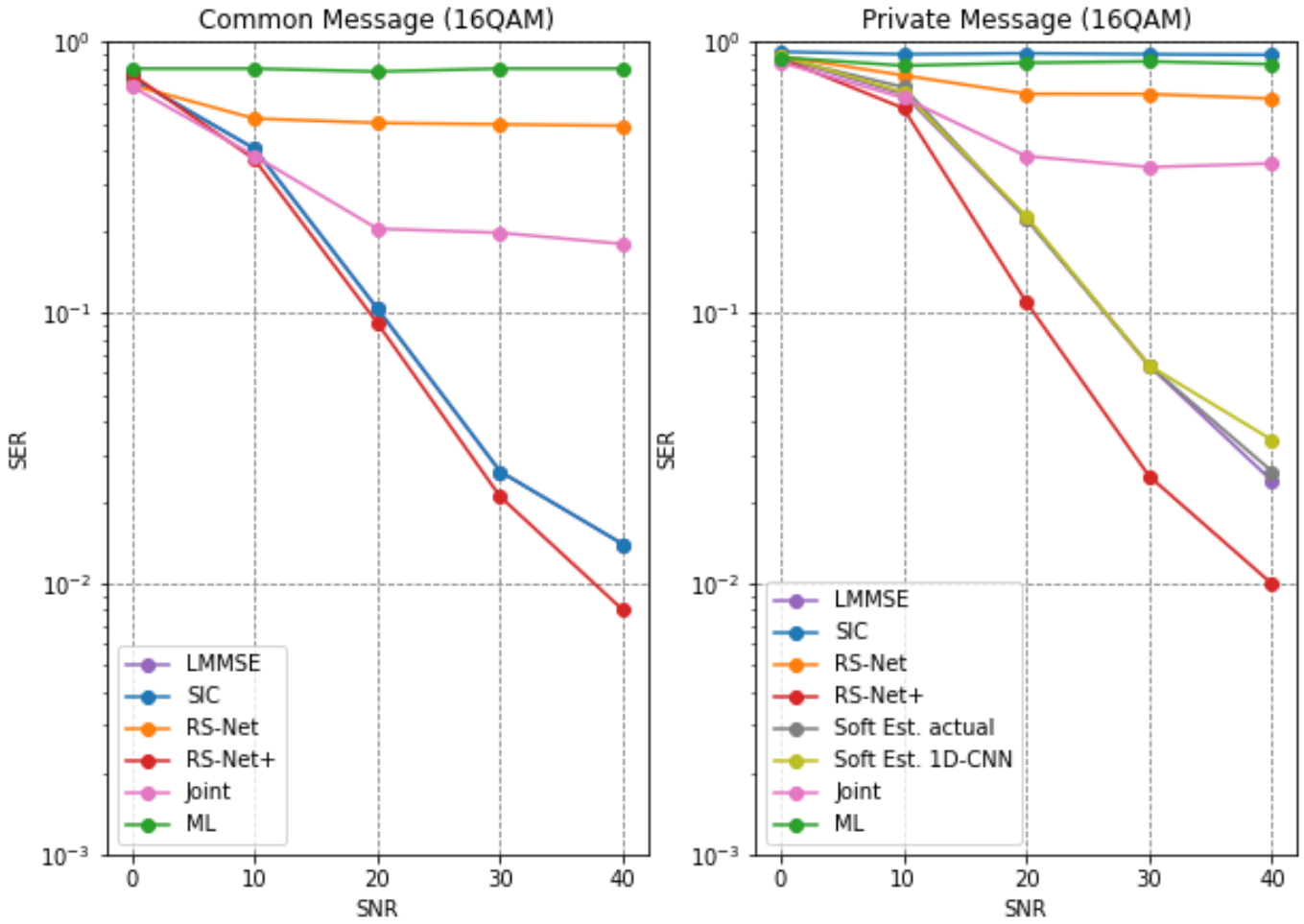
- RS-Net+ similar to SIC for both common and private SER.
- Joint-Net receiver crashes.

Non-linear channels (multi-antenna users, QPSK)



- ML exhaustive search receiver crashes.
- RS-Net+ and Joint-Net outperform ML, SIC.

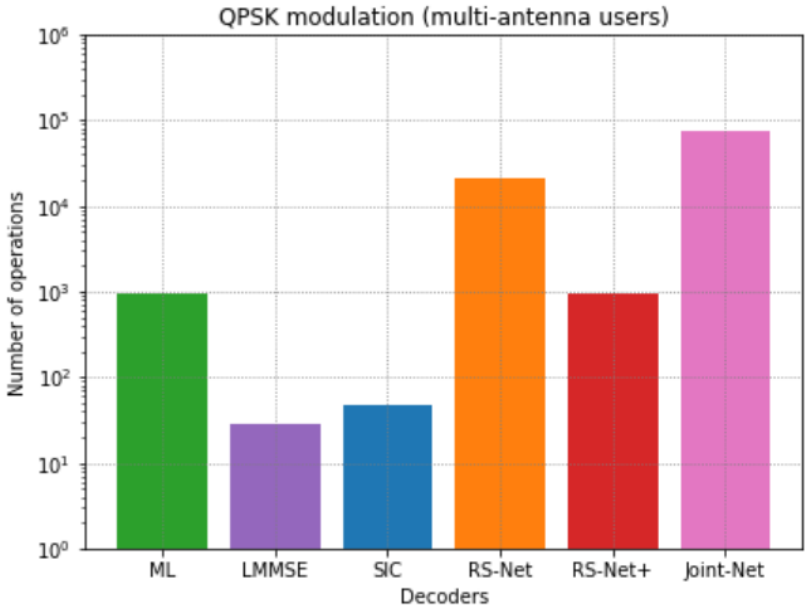
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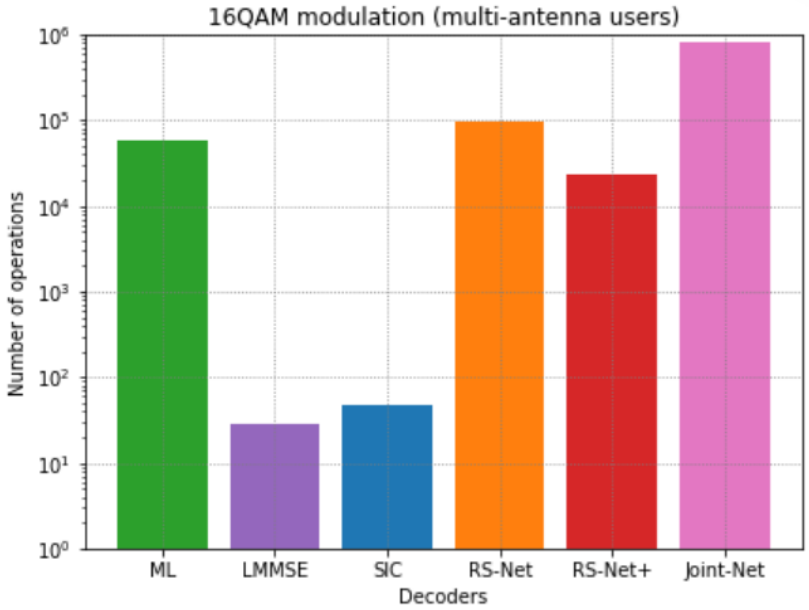


- ML and Joint-Net receivers crash.
- RS-Net+ outperforms SIC for private SER.

Computational complexity (Prediction Phase)

QPSK vs. 16QAM





- The computational complexity of RS-Net+ is lower than that of RS-Net and Joint-Net architectures.
- Data-driven approaches require higher number of operations than LMMSE and SIC receivers.

Pilot Overhead

NN Model	Number of pilot symbols
MAA[1]	960
TVL[2]	5000
JKM[3]	$10^{(5)}$
RCL[4]	1280
RS-Net	3

- RS-Net and Joint-Net architectures are trained for a given RSMA setting and channel conditions, thus do not require retraining for each realization.

[1] M. A. Aref and S. K. Jayaweera, "Deep learning-aided successive interference cancellation for MIMO-NOMA," GLOBECOM 2020

[2] T. V. Luong, N. Shlezinger, C. Xu, T. M. Hoang, Y. C. Eldar, and L. H. Hanzo, "Deep learning based successive interference cancellation for the non-orthogonal downlink," IEEE Transactions on VT 2022

[3] J. M. Kang, I. Kim, and C. J. Chun, "Deep learning-based MIMO-NOMA with imperfect SIC decoding," IEEE SJ 2020

[4] R. Cerna Loli, O. Dizdar, B. Clerckx, and C. Ling, "Model-based deep learning receiver design for rate-splitting multiple access," ArXiv 2022

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We have investigated the applicability of hybrid approaches against purely data-driven approaches in MIMO-RSMA setting

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Publications

Journal

(J1) **D. R. Kumar**, C. Antón-Haro and X. Mestre, "Deep Receiver Architectures for Robust MIMO Rate Splitting Multiple Access," in IEEE Transactions on Machine Learning in Communications and Networking, vol. 3, pp. 45-63, 2025, doi: 10.1109/TMLCN.2024.3513267.

Conferences

(C1) **D. R. Kumar**, C. Antón-Haro and X. Mestre, "RS-Net: Neural Network-Enhanced Receivers for MIMO Rate Splitting Multiple Access," 2023 IEEE Globecom Workshops (GC Wkshps), Kuala Lumpur, Malaysia, 2023, pp. 1964-1969, doi: 10.1109/GCWkshps58843.2023.10464670.

Other Publications

(O1) **D. R. Kumar**, C. Antón-Haro, and X. Mestre, "Applicability of Data-driven RSMA receivers: Linear and Non-linear channels," Proceedings of 7th Technical Meeting of INTERACT COST Action, Lisbon, Portugal, 2024.

(O2) **D. R. Kumar**, C. Antón-Haro, and X. Mestre, "Deep Learning-based Receivers for MIMO Rate Splitting Multiple Access," Proceedings of 5th Technical Meeting of INTERACT COST Action, Barcelona, Spain, 2023.

Outline

Introduction

Part I - Channel Estimation

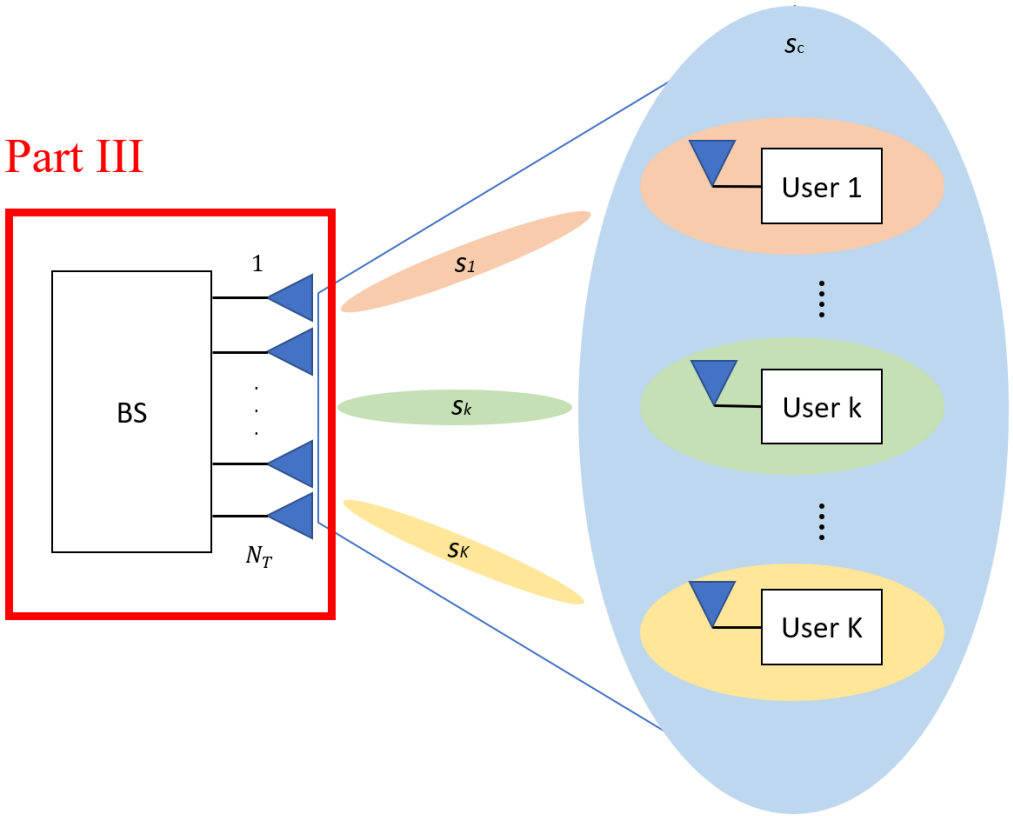
Part II - Symbol Detection

Part III - Precoder Optimization

Conclusions

Motivation

- Precoder optimization plays a key role in interference management. Conventional iterative schemes pose high computational complexity.
- ML approaches suitable for overcoming such limitations.



Signal Model

Transmitted signal in a MIMO-RSMA setting:

$$\mathbf{x} = \mathbf{PAs} = \alpha_c \mathbf{p}_c s_c + \sum_{k=1}^{K} \alpha_k \mathbf{p}_k s_k$$

SIC operation of the common stream $\Rightarrow$ SINRs at user k read:

$$\gamma_{c,k} = \frac{|\alpha_c \mathbf{h}_k^H \mathbf{p}_c|^2}{\sum_{q=1}^{K} |\alpha_q \mathbf{h}_k^H \mathbf{p}_q|^2 + \sigma_{n,k}^2}$$

$$\gamma_{p,k} = \frac{|\alpha_k \mathbf{h}_k^H \mathbf{p}_k|^2}{\sum_{q=1, q\neq k}^{K} |\alpha_q \mathbf{h}_k^H \mathbf{p}_q|^2 + \sigma_{n,k}^2}$$

- $\mathbf{P} = [\mathbf{p}_c, \mathbf{p}_1, \ldots, \mathbf{p}_k]^T \in \mathbb{C}^{N_T \times K+1}$ precoding matrix
- $\alpha_c, \alpha_k \in [0, 1]$: power allocation; such that $\alpha_c^2 + \sum_{k=1}^{K} \alpha_k^2 = 1$
- $\mathbf{h}_k \in \mathbb{C}^{1 \times M}$ is the channel of the k-th user

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Signal Model

RSMA Precoder Optimization - Maximizing the Achievable Sum-Rate

$$\max_{\mathbf{P}} \quad R_{\mathsf{sum}}(\mathbf{P}) = R_c + \sum_{k=1}^{K} R_k$$

$$\text{s.t.} \quad R_c \leq R_{c,k}; k = 1, \ldots, K$$

$$\mathrm{Tr}(\mathbf{P}\mathbf{P}^H) \leq P_t$$

State-of-the-art

RSMA Precoding Schemes

- RSMA Weighted Minimum Mean Square Error (WMMSE) precoding [5]
- Supervised Deep CNN-based precoding [6]
- Meta-learning MLP-based precoding [7]
- Self-supervised Transformers-based precoding [8]

[5] Y. Mao, B. Clerckx, and V.O.K. Li, "Rate-splitting multiple access for downlink communication systems: bridging, generalizing, and outperforming SDMA and NOMA," Eurasip Journal on Wireless Communications and Networking, 2018.

[6] Z. Wang, R. Ma, H. Shi, Z. Cai, L. Lin, and H. Guan, "Deep convolutional linear precoder neural network for rate splitting strategy of aerial computing networks," IEEE Transactions on Network Science and Engineering, 2024.

[7] R.C. Loli and B. Clerckx, "A meta-learning-based precoder optimization framework for rate-splitting multiple access,"IEEE Wireless Communications Letters,vol.13, pp.347–351, 2023.

[8] M. Wu, Z. Gao, C. Hu, and Z. Li, "Data-driven deep learning-based rate-splitting multiple access for FDD massive MIMO-OFDM systems with implicit CSI," 2023 IEEE 24th International Workshop on Signal Processing Advances in Wireless Communications (SPAWC), pp. 156-160, 2023.

Problem Formulation

Goal

To design the data-driven precoding policies in a self-supervised manner for 1-layer RSMA setting.

- Exploiting the structure of wireless graph via Graph Neural Network.
- Use of Shallow/Deep Multi Layer Perceptron, instead of Transformers.

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Self-supervised Precoder Optimization

Input: overall channel matrix $\mathbf{H} = [\mathbf{h}_1^T, \ldots, \mathbf{h}_K^T]^T \in \mathbb{C}^{K \times N_T}$
Output: precoding matrix $\widehat{\mathbf{P}}$

$$\widehat{\mathbf{P}} = f(\mathbf{H}; \theta)$$

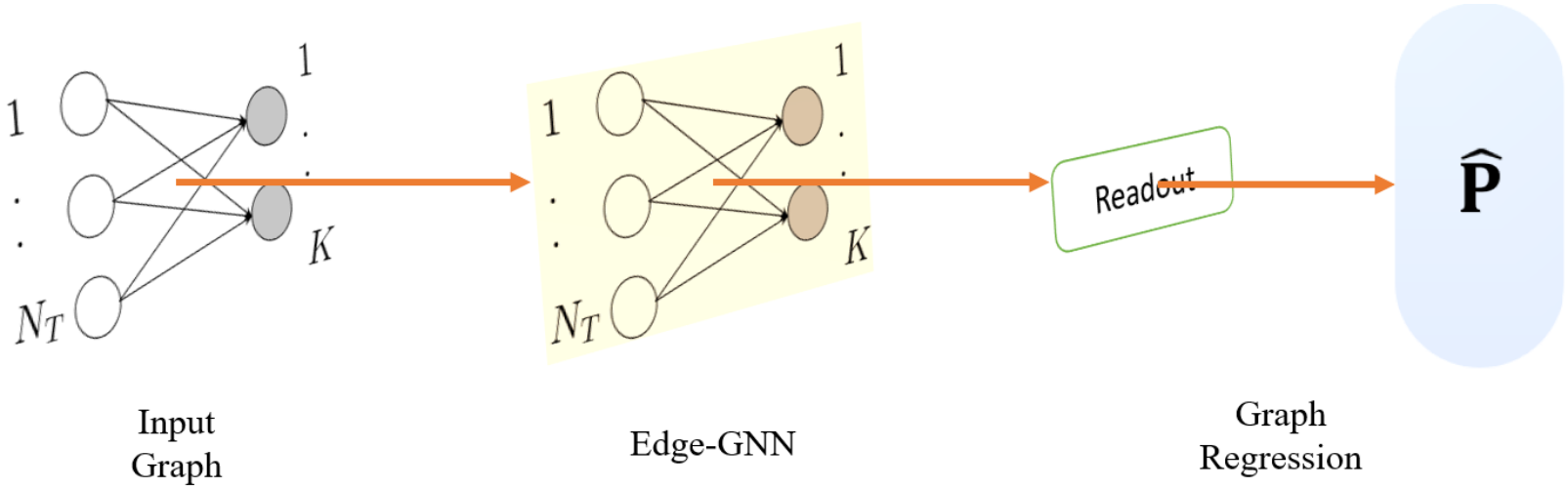
Loss Function: minimizing the negative of the sum rate

$$\theta^* = \arg\min_{\theta} -R_{\mathsf{sum}}(f(\mathbf{H}; \theta))$$

θ denotes the trainable parameters of the NN

NN Approach: GNN

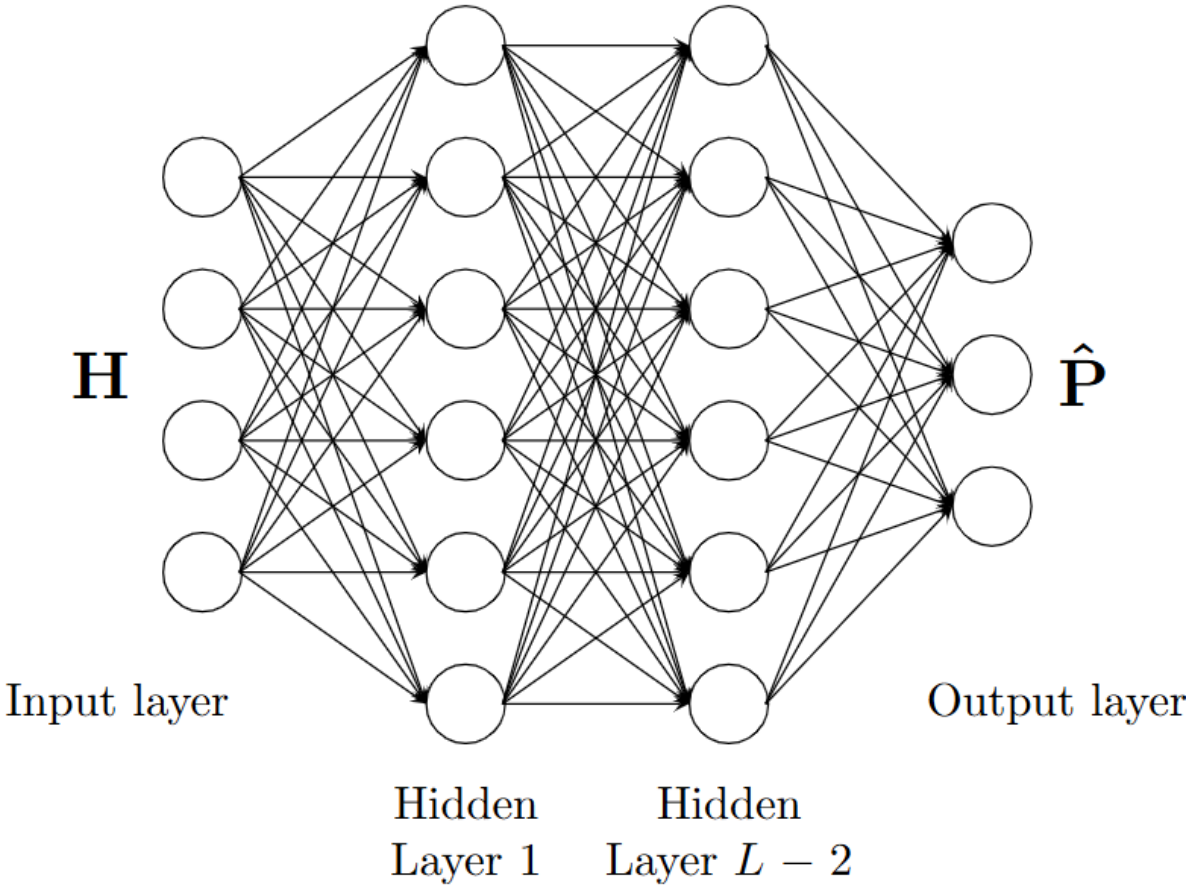
- Input: Bipartite graph with N_T transmit antennas and K users.



- The *edges* of input graph hold the channel coefficients.
- Edge-GNN performs message passing algorithm to update edge embeddings.

NN Approach: MLP

- Input: Row vector containinig the overall channel matrix.



Parameter set (unless otherwise stated)

Communication System:

- $K = 2, 8, 64$ single-antenna users.
- Highly-loaded and overloaded RSMA system.
- $(K + 1)$ streams served by N_T transmit antennas.
- MIMO channel: AWGN model, with i.i.d. entries.
- Performance measure: Achievable Sum-Rate.

Machine Learning schemes:

- GNN, shallow/deep MLP
- Benchmarks:
  - Zero-forcing (ZF) for private streams; Eigenvector with largest eigenvalue for the common stream (Part II)
  - Regularized Zero-forcing (RZF)
  - Weighted Minimum Mean Square Error (WMMSE) precoding

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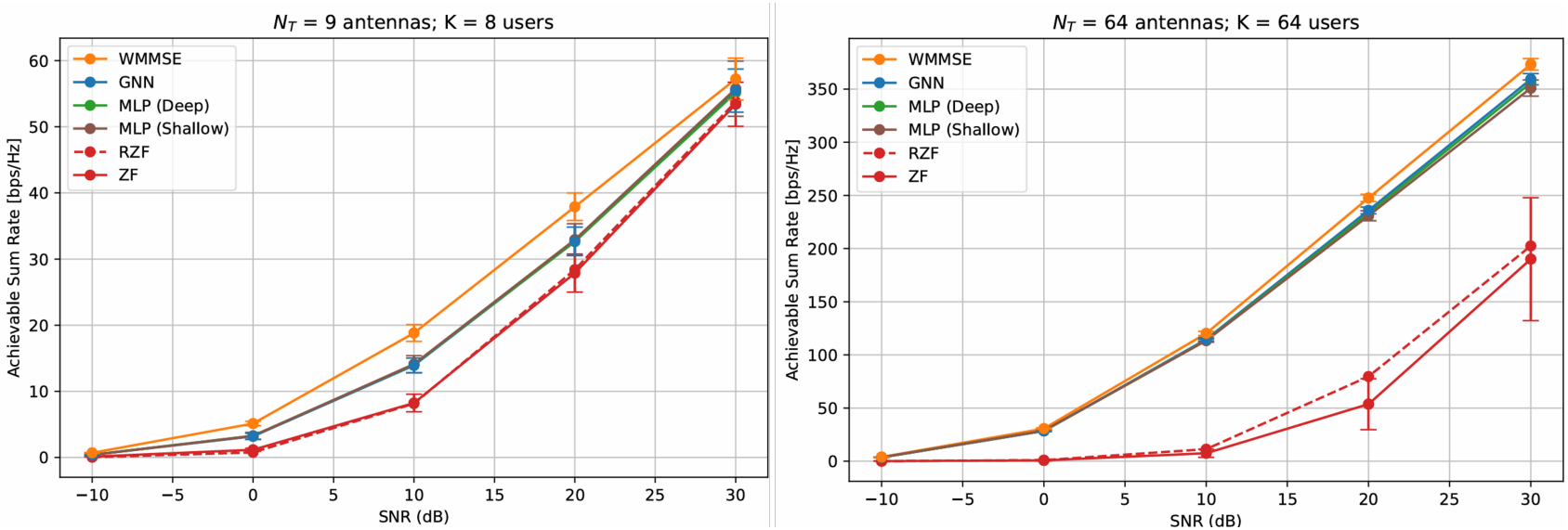
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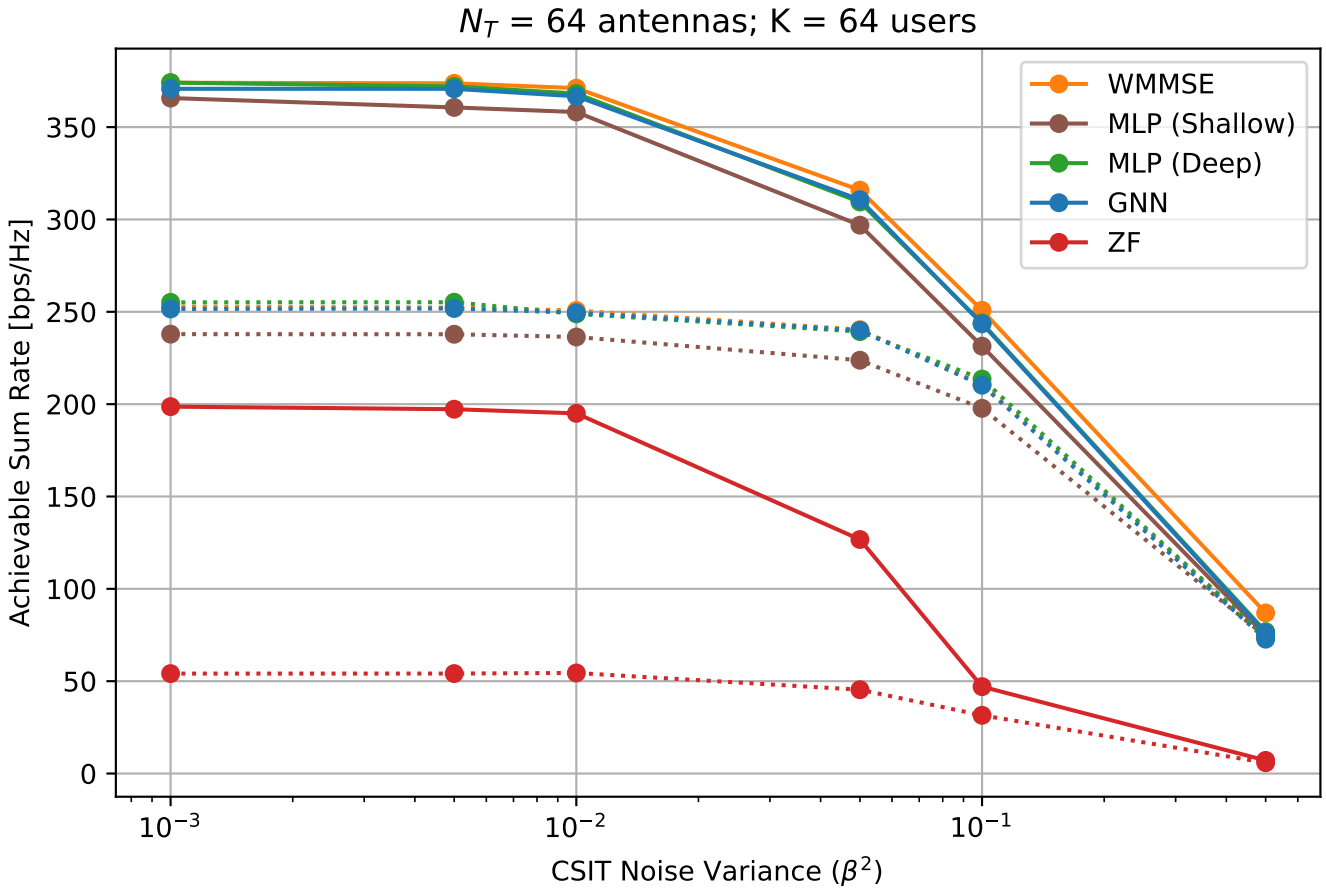
Performance of RSMA Precoding Policies

Achievable Sum-Rate



- In highly-loaded setting of 8 users, GNN and MLP perform better than ZF in the mid-SNR region.
- In overloaded RSMA setting of 64 users
  - GNN and MLP are similar to WMMSE.
  - Outperform ZF and RZF precoders.

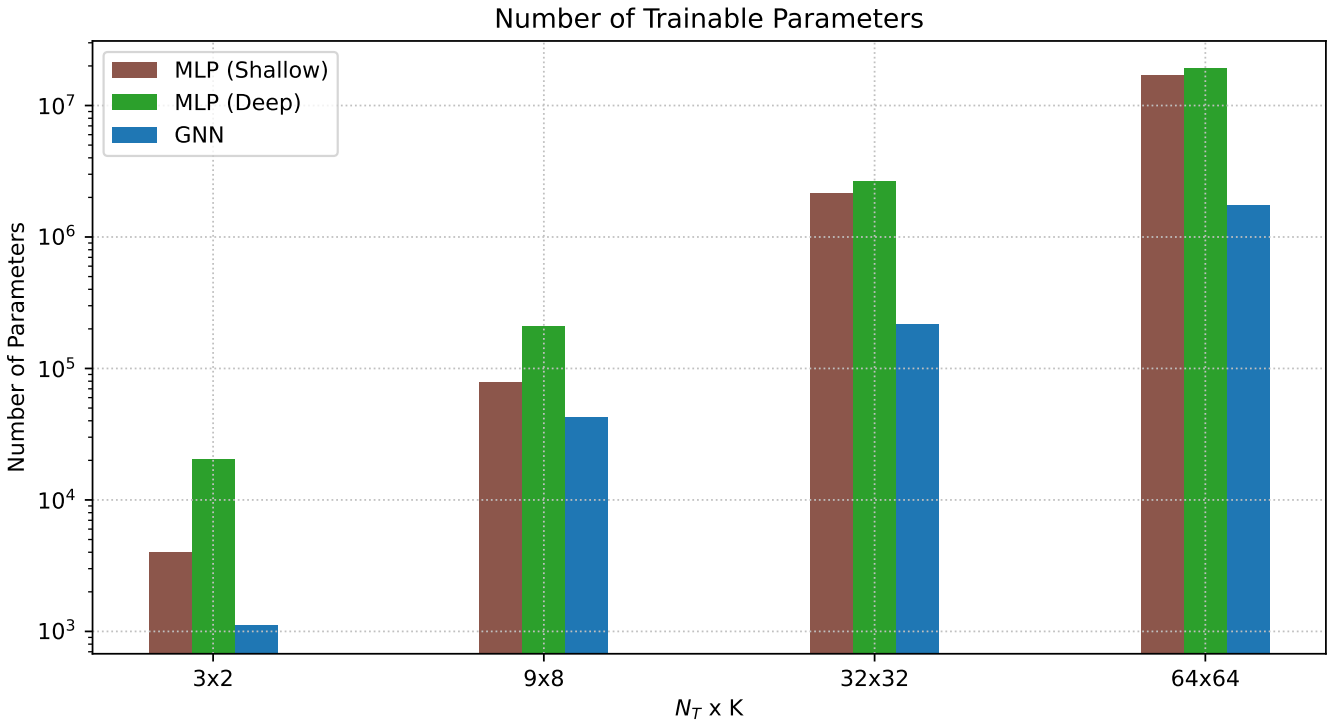
Impact of Channel Estimation Error



- GNN and MLP outperform ZF benchmarks for both 30dB and 20dB.

Computational Complexity Analysis

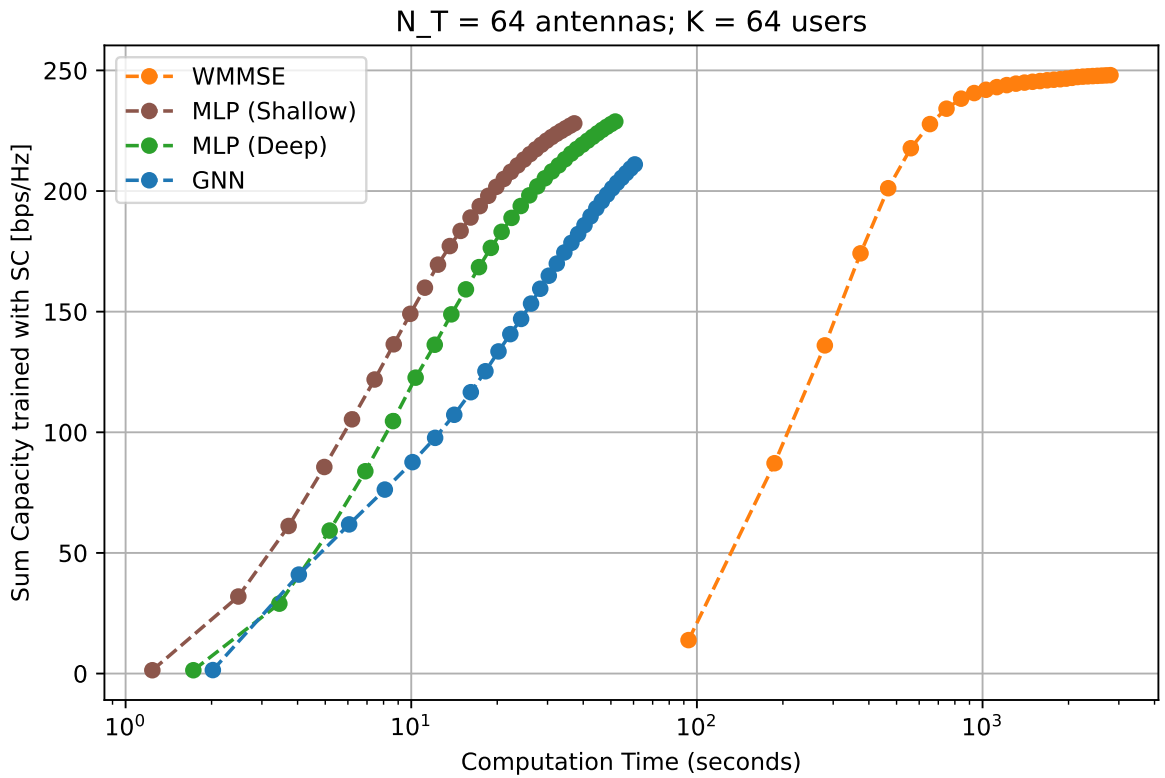
Total number of trainable parameters



- GNNs employ lesser number of parameters than MLPs.

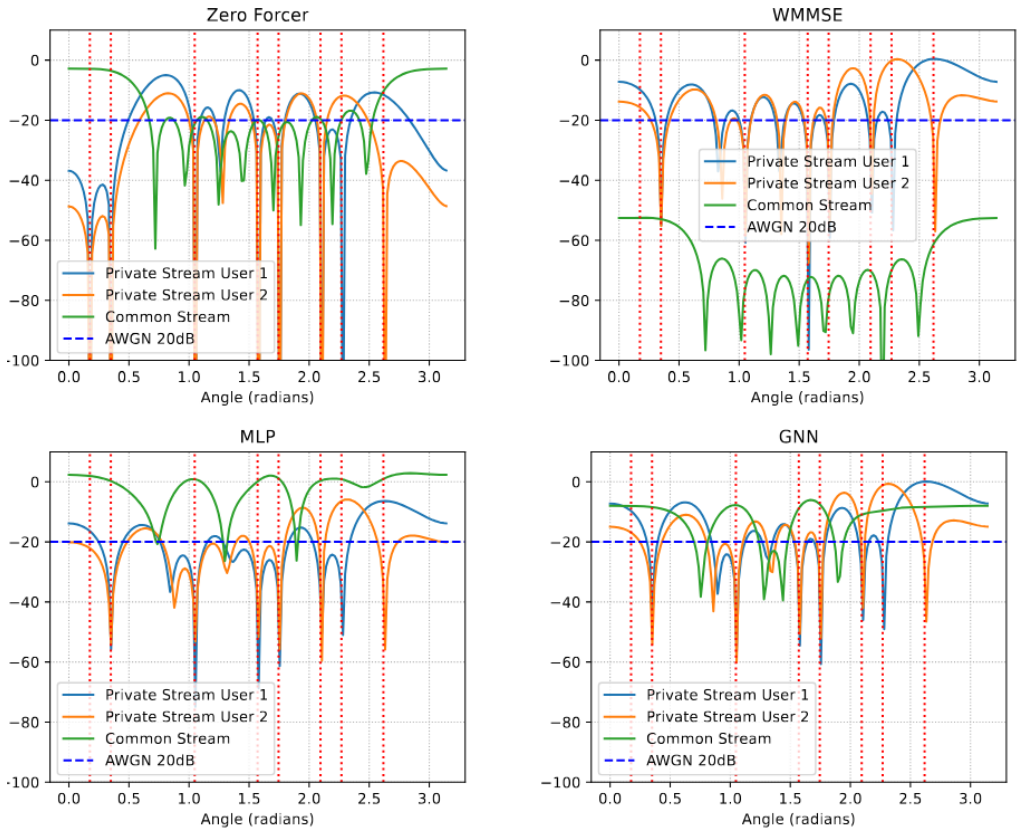
Computational Complexity Analysis

Computation Time



- GNNs and MLPs require less computation time than WMMSE.

Radiation Pattern Analysis



- Similar sum rates, but completely different precoding policies.
- WMMSE turns off the common stream, while the deep nulls in ZF do not mean higher sum rates.

Non-linear channels

- Non-linearities introduced by power amplifiers (PAs) at the transmitter side.
- The input of the PA in the BS is a bandpass signal which can be modeled as:

$$x(t) = A(t)e^{j\varphi(t)}$$

- Approximation of the non-linear distortion signal $\phi(\mathbf{x})$ as

$$\phi(\mathbf{x}) = \mathbf{B}\mathbf{x} + \mathbf{e}(\mathbf{x})$$

- Use of a third-order polynomial to model the non-linear PAs at each RF chain of the transmitter as follows

$$\mathbf{B}(\mathbf{P}) = \beta_1 \mathbf{I}_M + 2\beta_3 \text{diag}(\mathbf{P}\mathbf{P}^H)$$

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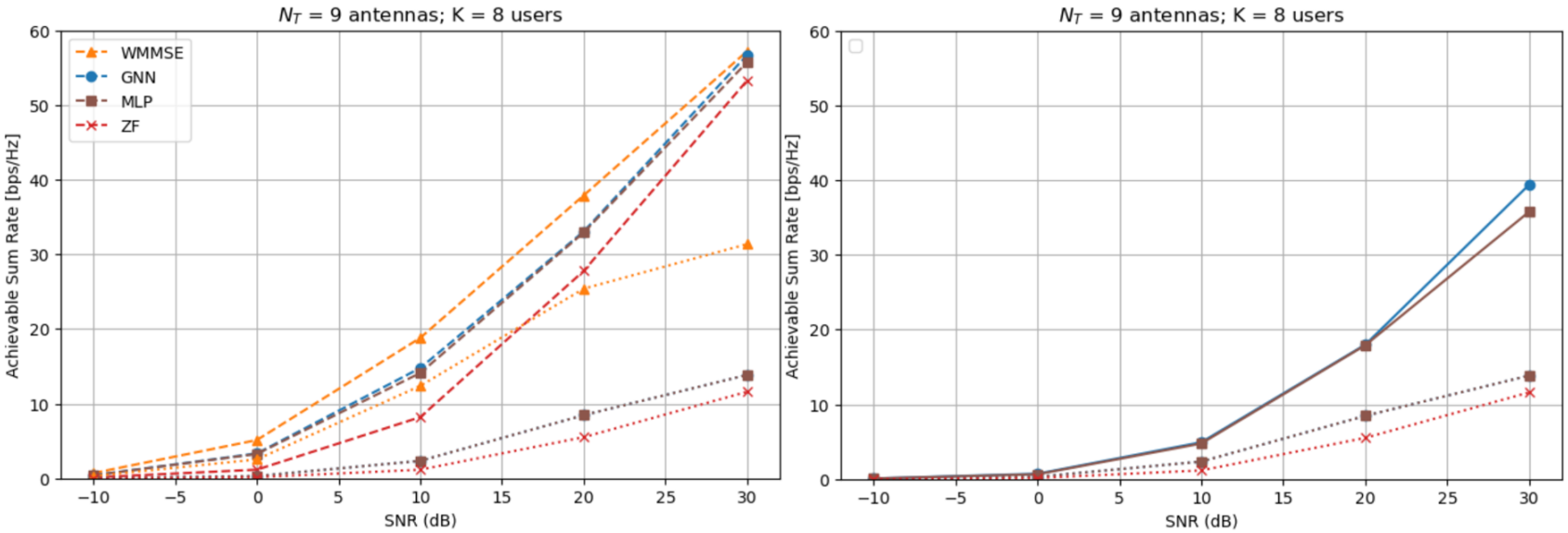
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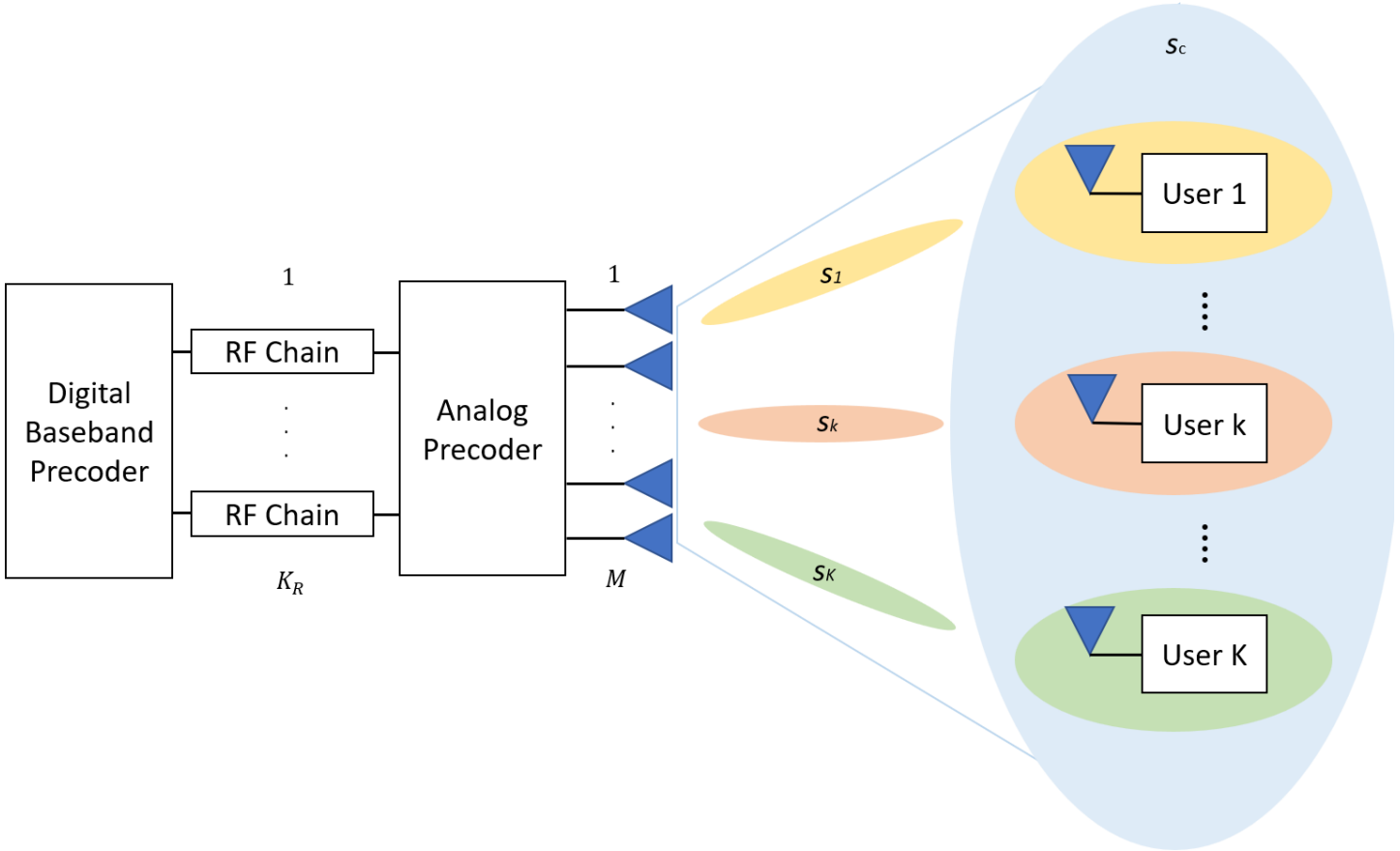
Achievable Sum-Rate



- (L sub-plot) Effect of model mismatch: performance drop of linear trained policies tested on non-linear channels.
- (R sub-plot) Inclusion of non-linear modeling in problem formulation: improved sum rate.

RSMA Hybrid Beamforming

- Hybrid analog-digital beamforming architectures offer reduced complexity and power consumption than fully digital beamforming.



Signal Model

Transmitted signal in a MIMO-RSMA setting:

$$\mathbf{x} = \alpha_c \mathbf{F}_{\mathsf{RF}} \mathbf{f}_{\mathsf{BB(c)}} s_c + \sum_{k=1}^{K} \alpha_k \mathbf{F}_{\mathsf{RF}} \mathbf{f}_{\mathsf{BB(k)}} s_k$$

SIC operation for the common stream $\Rightarrow$ SINRs at user k read:

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- $\mathbf{F}_{\mathsf{RF}} \in \mathbb{C}^{M \times K_{\mathsf{R}}}$ is Analog BF; $\mathbf{F}_{\mathsf{BB}} \in \mathbb{C}^{K_{\mathsf{R}} \times (K+1)}$ is Digital BF.
- $\alpha_c, \alpha_k \in [0, 1]$: power allocation; such that $\alpha_c^2 + \sum_{k=1}^{K} \alpha_k^2 = 1$
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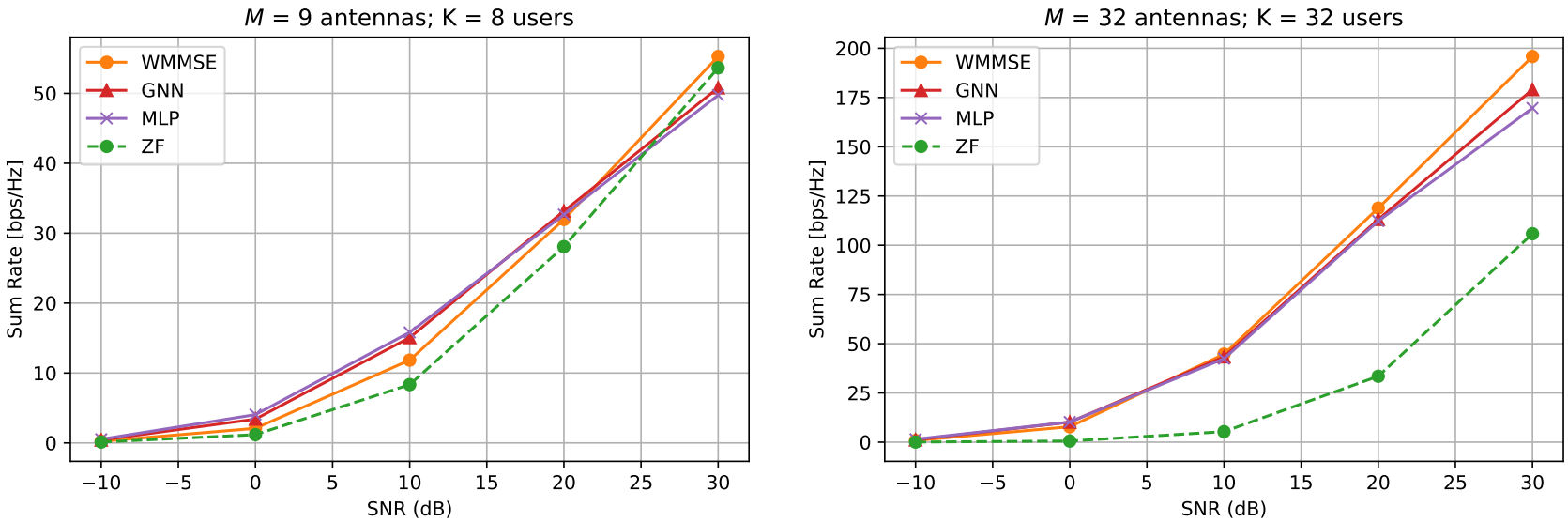
RSMA Precoder Optimization: maximize Achievable Sum-Rate

$$\begin{aligned} \underset{\mathbf{F}_{\mathsf{BB}}}{\text{maximize}} \quad & R_{\mathsf{sum}}(\mathbf{F}_{\mathsf{RF}}, \mathbf{F}_{\mathsf{BB}}) = R_c + \sum_{k=1}^{K} R_k \\ \text{subject to} \quad & R_c \leq R_{c,k}, \forall k \in K \\ & |\mathbf{F}_{\mathsf{RF}}| = 1/\sqrt{M} \\ & \mathsf{Tr}(\mathbf{F}_{\mathsf{BB}} \mathbf{F}_{\mathsf{BB}}^{H}) \leq P_t \end{aligned}$$

with $\mathbf{F}_{\mathsf{RF}}$ derived from the SVD of overall channel matrix $\mathbf{H}$.

Performance of RSMA Precoding Policies

Achievable Sum-Rate



- Highly-loaded setting (8 users): GNN and MLP outperform ZF in the mid-SNR region.
- Overloaded setting (32 users):
  - GNN and MLP similar to WMMSE.
  - Outperform ZF precoders.

Computational Complexity Analysis

Numerical comparison

Variable	ZF	MLP	GNN	WMMSE
Multiplications	$\mathcal{O}(K^3 + K^2M)$	$\mathcal{O}(E(MKN_l + N_l^2L))$	$\mathcal{O}(E(MKd_l^2L))$	$\mathcal{O}(ED(M^3 + K^2M^2))$
Additions	$\mathcal{O}(K^3 + K^2M)$	$\mathcal{O}(EN_lL)$	$\mathcal{O}(E(MKd_l^2L + M^2Kd_lL + MK^2d_lL))$	$\mathcal{O}(ED(M^3 + K^2M^2))$

- The self-supervised training of GNN and MLP is done for a number of epochs (E) which includes operations in forward and backward pass.
- WMMSE: complexity of the convex optimization (CVX) subroutine increases dramatically at the rate of $\mathcal{O}(M^3)$.
  - CVX subroutine is an iterative solver that is called D times for each epoch; accounts for (ED) number of iterations.
- GNNs and MLPs require less computation operations than WMMSE.

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Conclusions

We have investigated the applicability of self-supervised graph-based and shallow MLP approaches for RSMA precoding

- GNNs and MLPs require less computation time than WMMSE precoding scheme.
- GNN and MLP outperform sub-optimal ZF by several orders of magnitude.
- GNNs prove to be a scalable solution requiring lesser trainable parameters.
- GNN performance is much closer to that of MLP, without the need for exhaustive hyperparameter tuning.

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(C1) **D. R. Kumar**, T. Feys, C. Antón-Haro, F. Rottenberg, and X. Mestre, "Graph Neural Networks for Hybrid Beamforming in MIMO Rate Splitting Multiple Access," 2025 IEEE International Conference on Machine Learning for Communication and Networking, Barcelona, Spain, 2025.

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Other Publications

(O1) **D. R. Kumar**, T. Feys, C. Antón-Haro, F. Rottenberg, and X. Mestre, "Hybrid Beamforming in MIMO Rate Splitting Multiple Access via Graph Neural Networks," Proceedings of 10th Technical Meeting of INTERACT COST Action, Dublin, Ireland, 2025.

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PhD Conclusions

- The choice of neural network architecture and input feature representation affects performance and computational complexity of AI/ML solutions.
  - 1D-CNN and GNN for MIMO channel estimation (Part I).
  - GNN for RSMA precoder optimization (Part III).
- Hybrid approaches over black-box solutions offer better performance under challenging settings.
  - RS-Net+ offers reliable symbol detection in 16QAM and non-linear channels (Part II).
- The ability of representing signal processing tasks on a graph offers scalable solutions giving reliable performance.
  - GNN for large MIMO channel estimation and RSMA precoder optimization (fully digital and hybrid beamforming).

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Future Research Lines

- Investigate the design of GNN architectures for user grouping in MIMO-RSMA settings.
  - To address the computational complexity in jointly optimizing user grouping and decoding order in 2-layer RS and Hierarchical RS.
- Applicability of transfer learning and split learning for RSMA precoding.
  - To increase the convergence rate of the data-driven precoding schemes.
- Design and implementation of reinforcement learning solution for analog precoder selection.
  - To enable adaptive and dynamic decision-making.
  - Study the impact of digitization/quantization of analog phase values on precoder optimization.

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Involvement at ETSI MAT 2025

- Representation of CTTC's involvement at ETSI Industry Specification Group (ISG) on Multiple Access Technologies (MAT) in 2025. The members are from BBC, ICL, CTTC, VIAVI Solutions, among others.

Thank You!

To the friends who became family
and to my family becoming friendly.

*****

Dheeraj Raja Kumar
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CTTC, Castelldefels, Barcelona, Spain

September 25, 2025